

Do Financial Regulations Shape the Functioning of Financial Institutions' Risk Management in Asset-Backed Securities Investment?

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We show that installing stronger risk management into financial institutions—a proposal widely discussed following the 2008 financial crisis—is insufficient to constrain institutions' exposure to investment with lurking risk, such as asset-backed securities (ABS). Regulations affect the functioning of risk management: risk management constrains institutions' exposure to risky ABS when they face mark-to-market reporting combined with capital requirements; however, this role is considerably weaker when capital requirements are combined with historical cost accounting. We find suggestive evidence that financial regulations affect risk management functions through promoting risk managers' efforts in uncovering ABS risk and curbing executives' incentives to take excessive risk. (*JEL* G20, G22, G28, G32)

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“Hundreds of billions of dollars in losses in mortgages and mortgage-related securities shook markets as well as financial institutions that had significant exposures to those mortgages. ... We conclude dramatic failures of corporate governance and risk management at many systemically important financial institutions were a key cause of this crisis.”

—Financial Crisis Inquiry Commission (FCIC, 2011)

Before the 2008 financial crisis, the securitization market built up considerable credit risk due to agency problems in the securitization process (e.g., Keys et al. 2010; Purnanandam 2011; Jiang, Nelson, and Vytacil 2013), inefficiencies in contracting (e.g., DeMarzo and Duffie 1999; Agarwal et al. 2011), and perhaps more strikingly, information misreporting (e.g., Jiang, Nelson, and Vytacil 2014; Griffin and Maturana 2016). However, this risk was masked by the favorable risk assessments of securitized products *ex ante*, such as high credit ratings (Securities and Exchange Commission [SEC] 2008). When such “lurking” risk eventually unveiled itself in the crisis, it intoxicated many asset-backed securities (ABS) (i.e., mortgage-backed securities [MBS], collateralized debt obligations [CDO], and other assets-backed securities) and depleted the capital of financial institutions (Brunnermeier 2009; Erel, Nadauld, and Stulz 2014).

The securitization market has since been under the spotlight of regulators (see the epigraph). Growing attention is given to how regulatory policies may help prevent financial institutions from taking excessive risk. At the center is the proposal to install stronger risk management in financial institutions. It is believed that by setting up enterprise-wide risk management (ERM) and dedicated risk committees, institutions can strengthen risk controls and, hence, become more effective in constraining their exposure to risky investment, particularly ABS (FCIC 2011; the Dodd-Frank Act, Section 165).¹ In this paper, we take a closer look at this perspective. We examine whether risk management’s function in constraining institutions’ exposure to ABS before the 2008 financial crisis hinges on the financial regulations they face—the combination of capital requirements and regulatory reporting rules (mark-to-market or historical-cost accounting).

To understand why financial regulations could affect the function of risk management, one needs to first understand the role of risk management in

¹ The 2010 Dodd-Frank Act requires financial institutions to install stronger risk management. Under the act, institutions should implement enterprise-wide risk management, and form dedicated risk committees that include qualified individuals to help identify, assess, and manage risk exposures, and independently oversee the enterprise-wide risk management practices.

constraining ABS investment. Risk management aims to help institutions conserve economic capital and keep them away from investments that appear safe but contain excessive risk (Nocco and Stulz 2006). Strong risk management provides necessary infrastructure to achieve this goal: by appointing risk managers responsible for risk identification, measurement, reporting, and treatment, and by designating risk committees to oversee firms' risk profile and investment portfolios, risk management lays the foundation to identify ABS risk. However, as pointed out in Ellul and Yerramilli (2013), this foundation alone may not be sufficient to warrant effective risk management. Two complementary conditions are needed.

First, risk managers need to look beyond apparent risk assessments so as to identify the underlying, but often lurking, risk of ABS. Second, executives need to acknowledge and treat the identified risk so that risk controls can be ultimately enforced. Both conditions, however, are largely compromised before the 2008 crisis. First, the complex ABS structure and possible information falsification have obscured the underlying risk of ABS (Erel, Nadauld, and Stulz 2014; Ghent, Torous, and Valkanov 2019). To uncover such risk, risk managers need to put in more due diligence through probing ABS fundamentals in registration statements, scrutinizing the disclosure quality of ABS prospectuses (Liang et al. 2017), and/or examining executives' stock selling of banks that underwrite ABS.² Second, the favorable ABS risk assessments provided by credit ratings foster an agency problem among executives and traders with high-powered compensation schemes: they are incentivized to build up ABS exposure despite the high underlying risk. After all, ABS often provide higher yields than other securities (e.g., corporate bonds), and their "safe" appearance is unlikely to raise red flags among investors or regulators (Calomiris 2008; Senior Supervisors Group 2008; Erel, Nadauld, and Stulz 2014).

These situations, as highlighted in Shleifer (2011), are where financial regulations come to play an important role. Financial institutions, such as banks and insurance companies, are subject to capital requirements and regulatory reporting rules: mark-to-market (MTM) versus historical-cost accounting (HCA). Capital requirements aim to constrain institutions from excessive risk raking. These requirements typically use a capital adequacy metric to determine whether a company's capital level is adequate given the degree of risk that the firm takes. The riskier an institution's investment, the more statutory capital it is required to hold. On the other hand, regulatory reporting rules determine whether institutions should immediately recognize asset value erosions (under MTM), or have the flexibility to avoid doing so by booking impaired assets at historical costs (under HCA).

² Executives of ABS underwriters are found to have exploited their private information on nonperforming securitization income or securitization-related recourse risks to capture short-term profit ahead of the crisis (Bebchuk, Cohen, and Spamann 2010; Ryan, Tucker, and Zhou 2015; Cziraki 2018; Kolasinski and Yang 2018).

Ellul et al. (2014, 2015) show that the combination of these regulations distinctly shapes institutions' trading strategies on ABS. Following large rating downgrades of ABS, institutions under HCA continue to hold downgraded ABS at historical costs, avoiding recognizing the value decline of these securities. To make up for the lowered capital adequacy ratio due to increased risk-weighted assets, they sell corporate and government bonds with the largest unrealized gains. In contrast, institutions under MTM have to immediately recognize value deterioration of troubled ABS. The resulting losses lower their capital adequacy ratio. To fulfill capital requirements, they fire-sell downgraded ABS—despite severely depressed prices amid massive downgrades; doing so allows them to swap risky ABS for cash, thereby reducing risk-weighted assets and required capital.

Therefore, unlike HCA, MTM—when combined with capital requirements—forces institutions to promptly realize ABS losses at the trough of market value. This anticipated ramification can in turn promote institutions' risk recognition by (i) pressuring risk managers to perform greater due diligence *ex ante* to discover the underlying ABS risk, and (ii) curbing executive incentives to pursue profits by building up excessive ABS investment. Ultimately, the combination of MTM and capital requirements reinforces the two conditions needed for risk management, engendering more effective risk controls than HCA and capital requirements.

We empirically examine these different roles of financial regulations. Although such roles may be observed among various financial institutions, including banks and insurance companies, we use U.S. insurers as a laboratory for a few reasons. First, there are two major types of insurers: property and casualty (P&C) insurers and life insurers. While both types face common capital requirements, they are governed by different regulatory reporting rules: P&C insurers follow MTM, and life insurers typically follow HCA.³ Because individual insurers have little discretion over which reporting rule they should comply with, we can naturally use insurer types to identify reasonably exogenous variations in financial regulations. Moreover, state-level insurance codes often differ in the extent of market value recognition. We can therefore further examine variations in market value recognition (i.e., how close the reporting rules are to MTM) within life insurers. This analysis helps us mute any confounding differences between P&C and life insurers.

We collect detailed information on ABS investment of U.S. insurers. Insurers have emerged as one of the largest investor groups in the ABS market. They collectively hold approximately U.S.\$500 billion of ABS as of 2007, exceeding the holding amount of U.S. bank-holding companies (BHCs). While banks disclose ABS holding at the aggregate level (Erel, Nadauld, and Stulz 2014), insurers are required by the National Association of Insurance Commissioners

³ See Section 1 for details on capital requirements and regulatory reporting rules for the two types of insurers.

(NAIC) to disclose ABS investment at the security level. This information enables us to track the risk assessments of ABS (such as credit rating changes) over time. In our analyses, we use this information to study whether strong risk management helps identify the “lurking” risk of ABS—risk that is masked by favorable assessments *ex ante* but revealed in *ex post* rating downgrades. Following Ellul and Yerramilli (2013), we develop a risk management index to gauge the strength of an institution’s risk oversight (details in Section 2.2).

We document a few important findings. First, strong risk management not only reduces insurers’ level of ABS investment at the onset of the financial crisis (2005–2007), but also leads them to invest in ABS that have higher quality. These ABS are less likely to be substantially downgraded from a AAA rating to a speculative grade *ex post*; they experience a smaller decline in credit quality; and they are less likely to be underwritten by “reputable” banks (and hence are disguised as high-quality securities), but ultimately suffer the most capital losses (Griffin, Lowery, and Saretto 2014). Therefore, risk management helps financial institutions identify and reduce ABS exposures that appear safe but end up in trouble.

However, this function of risk management is only effective among insurers that are subject to MTM and capital requirements. It is considerably weaker for those governed by HCA. Financial regulations therefore significantly affect the efficacy of risk management in constraining institutions’ ABS investment exposure. We further explore the channels through which financial regulations affect the functioning of risk management. We find suggestive evidence that MTM, combined with capital requirements, helps promote greater due diligence from risk managers in discovering the underlying ABS risk, and curb executives’ incentives to build up excessive ABS exposure for profits.

Amid ongoing efforts to constrain financial institutions’ risk taking, our findings suggest that installing risk management alone might be insufficient. Regulators should consider the benefit of external regulations—namely the prompt recognition of investment losses enforced by MTM and capital requirements—in complementing institutions’ internal risk management. This consideration is particularly salient because institutions are not often incentivized to voluntarily recognize investment losses promptly, and so cannot internalize the benefits of these external regulations (FCIC 2011).

Our main analyses employ P&C versus life insurers to identify variations in financial regulations. We recognize that the two types of insurers may differ along other dimensions such as firm characteristics and business natures. To mitigate the concern that the differential effects of risk management on ABS investment may be simply driven by these confounding differences, we take several complementary approaches. First, we employ propensity score matching to derive a set of P&C and life insurers that are comparable in key firm characteristics. To the extent that confounding differences between the two types of insurers are reflected in observable characteristics, this analysis helps us isolate the role of financial regulations in affecting the efficacy of

risk management. Second, we take advantage of the variations in market value recognition within the life insurance sector. In the United States, insurance regulations are implemented at the state level, and this practice grants a state insurance commissioner discretion when applying the NAIC-model rules to insurers domiciled in that state. This discretion generates variations in the recognition of market value and, hence, determines how close a state's insurance rule is to the MTM rule within the life insurance sector. We show that risk management of life insurers domiciled in states with a higher tendency of market value recognition (analogous to MTM combined with capital requirements) is more effective in constraining ABS exposure than that of life insurers domiciled in states with a lower tendency of market value recognition. This analysis therefore mutes the confounding differences between P&C and life insurers that may not be fully captured by the propensity score matching. Third, we exploit changes in firms' risk management strength following an exogenous event (the 9/11 terrorist attacks) and examine whether our findings withstand a difference-in-differences (DID) setting that is more resilient to unobserved confounding factors. We confirm our findings.

Two points related to our findings are worth noting. First, our findings do not imply that external financial regulations can substitute institutions' internal risk management. Internal risk management provides necessary infrastructure for risk identification, measurement, and treatment; therefore, it constitutes an irreplaceable part of institutions' risk controls. Our findings instead point to the complementarity between external regulations and internal risk management.

Second, our finding that life insurers, especially the ones with a lower tendency of market value recognition, exhibit less effective risk controls in constraining ABS risk does not mean that they initially install ineffective risk management policies. In fact, their risk management policy may well intend to identify and treat risk in the same way as that of P&C insurers. Our findings instead suggest that the fulfillment of such a policy may be hampered by frictions these life insurers face, in particular, risk managers' weakened incentives to perform due diligence, or executives' higher temptation to neglect identified risk under HCA. These frictions arise because under HCA, insurers' reported income and regulatory capital become less sensitive to the market value changes of security holdings and therefore are largely shielded from potential value declines in ABS. Such a shield affords risk managers some latitude to become lax in performing costly due diligence and enables executives with high-powered compensation to aggressively build up ABS exposures for their high yields, given that potential losses will not be promptly reflected in firms' reported income or adversely affect executive compensation. Eventually, these frictions make risk management less effective in constraining ABS risk under HCA.

From this perspective, our findings are analogous to those in the literature on credit rating agencies (e.g., Bolton, Freixas, and Shapiro 2012; Griffin and Tang 2012; Xia 2014). This literature shows that while rating agencies set the policy to assign ratings to reflect securities' underlying credit

risk, such a policy is often hampered by frictions in the rating process—that is, biased incentives to assign issuer-friendly ratings. As a result, ratings often become ineffective in revealing underlying credit risk. Our findings highlight the necessity for policy makers to consider both the installation of sound risk-management policy as well as the fulfillment of the policy that may be affected by external financial regulations.

Lastly, we explore the value implications of risk management. Specifically, we examine whether strong risk management—by constraining the exposure to ABS—eventually affects insurers' investment performance and tail risk, and whether this effect depends on the financial regulations they face. The results are in line with our previous findings. Among P&C insurers, strong risk management mitigates the adverse effect of ABS investment on overall investment performance and tail risk. As expected, this effect is considerably weaker for life insurers.

Our paper contributes to the literature in two ways. First, amid the growing regulatory discussions on whether risk management fails to constrain financial institutions' ABS exposure, two important studies, Ellul and Yerramilli (2013) and Erel, Nadauld, and Stulz (2014), examine the effect of bank risk management on the level of ABS investment. We add to these studies by further examining risk management's effect on the quality of ABS investment. This is important given that the core function of risk management is not to avoid risk altogether, but to avoid value-destroying risk—the “lurking” risk of ABS in our context (Stulz 2014).

Second, we document that the functioning of risk management hinges on external financial regulations, and it is only effective among P&C insurers and life insurers domiciled in states with a higher tendency of recognizing the MTM rule. This result extends the findings in Ellul et al. (2014, 2015), which show that regulatory reporting rules, in interaction with capital requirements, affect financial institutions' ABS trading behavior. This behavior ultimately engenders price pressure on corporate bonds, even though they are not directly related to ABS. This negative externality should be considered when reforming the regulations governing financial institutions. Our results highlight another externality of regulations: HCA when combined with capital requirements may hinder the efficacy of risk management. This implication also adds to the recent debates on the pros and cons of the MTM and HCA reporting systems (Plantin, Sapra, and Shin 2008; Heaton, Lucas, and McDonald 2010; Laux and Leuz 2010; Ellul et al. 2015).

1. Institutional Background and Hypotheses Development

The ABS market witnesses a significant rise in credit risk before the onset of the 2008 financial crisis. Figure 1 illustrates this point by showing the percentage of ABS that are downgraded by Standard and Poor's (S&P) in the crisis (i.e., as of 2009), as well as the average magnitudes of rating changes. Figure 1

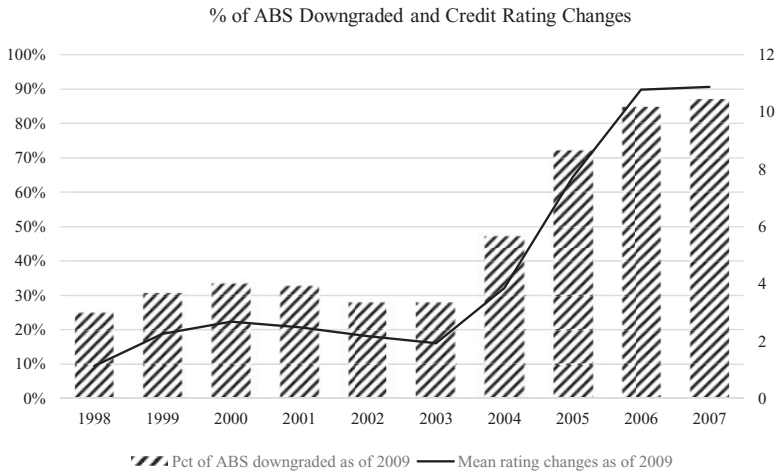


Figure 1
Percentage of ABS downgraded and rating changes

This figure presents the percentage of asset-backed securities (ABS) that were downgraded by Standard and Poor's (S&P) during the financial crisis (i.e., as of 2009), as well as the average magnitudes of ABS rating changes. ABS include MBS, CDO, and other assets-backed securities. The figure covers ABS issued from 1998 to 2007. The columns represent the percentage of ABS issued in a year that is downgraded as of 2009 (the left y-axis), and the solid line shows the magnitudes of average rating changes as of 2009 for ABS issued in a year (the right y-axis). We assign a numerical value to each S&P rating (1 = AAA, 2 = AA+, 3 = AA, 4 = AA- ..., 22 = D) and calculate rating changes as the difference between the rating of an ABS at the end of 2009 and the initial rating when the ABS is issued. Positive (negative) rating changes indicate rating downgrades (upgrades).

covers ABS issued between 1998 and 2007. We start from 1998 because the issuance of ABS before 1998 is relatively small. The bars in Figure 1 represent the percentage of downgrades among the ABS issued each year (the left y-axis), and the solid line shows the magnitudes of average rating changes for the ABS issued each year (the right y-axis). We assign a numerical value to each S&P rating (1 = AAA, 2 = AA+, 3 = AA, 4 = AA- ..., 22 = D) and calculate rating changes as the difference between the ABS rating as of 2009 and the initial rating when an asset-backed security is issued. Positive (negative) rating changes indicate rating downgrades (upgrades).

Figure 1 shows that there is a surge in ABS rating downgrades from 2005 to 2007. While ABS issued before 2005 experience only a moderate rating downgrade in the crisis (with an average downgrade not exceeding 4 notches as of 2009), the ones issued between 2005 and 2007 suffer from massive downgrades, with a magnitude of up to 11 notches. According to Griffin and Tang (2012), most of these downgraded ABS bear a high rating (often AAA) ex ante, and hence, the credit risk of the ABS during 2005–2007 is largely concealed by favorable risk assessments until the crisis unfolds. Such “lurking” risk is further demonstrated in Table 1, which presents rating transitions of ABS held by insurers. Table 1 shows that among AAA-rated ABS as of 2007, almost 40% of them are downgraded as of 2009, and over 25% of them are

Table 1
Rating downgrades of ABS held by U.S. insurance companies

	AAA	AA	A	BBB	BB	B	CCC	CC	C and below	# Total ABS	# Downgrades	% Downgrades	# Downgrades to speculative	% Downgrades to speculative
AAA	7,231	484	481	538	482	566	1,762	136	81	11,761	4,530	38.52%	3,027	25.74%
AA	36	1,644	185	177	203	232	364	120	83	3,044	1,364	44.81%	1,002	32.92%
A	12	41	1,141	211	204	316	310	179	131	2,545	1,351	53.08%	1,140	44.79%
BBB	11	0	44	724	137	197	210	187	150	1,660	881	53.07%	881	53.07%
BB			2	7		30	40	45	80	341	195	57.18%		
B					1	62	27	23	68	181	118	65.19%		
CCC							11	18	56	85	74	87.06%		
CC								1		1	0	0.00%		
Total	7,290	2,169	1,853	1,657	1,164	1,403	2,724	709	649	19,618	8,513	43.39%	6,050	30.84%

This table shows the credit rating downgrades for asset-backed securities (ABS) held by all U.S. insurance companies. Ratings in the left column indicate ratings of ABS as of the end of 2007, and ratings in the top row indicate ratings as at the end of 2009. # Total ABS is the number of ABS held by insurance companies. # Downgrades is the total number of ABS that are downgraded as at the end of 2009. % Downgrades is the ratio of # Downgrades to # Total ABS. # Downgrades to speculative is the total number of ABS that are downgraded to a speculative grade (below BBB). % Downgrades to speculative is the ratio of # Downgrades to speculative to # Total ABS.

downgraded to speculative grade. Moreover, almost a third (30.8%) of ABS rated as investment grade before the crisis ended up in the speculative grade.

Taken together, these observations point to a wedge between ex ante risk assessments and the underlying quality of ABS, particularly during 2005–2007. This wedge is where risk management can come to play an important role. Risk management ought to identify the underlying risk of ABS investment and keep institutions away from securities that appear safe but contain excessive risk. We argue that whether this goal can be effectively achieved depends on the financial regulations faced by financial institutions—specifically, capital requirements and regulatory reporting rules (MTM or HCA).

In the United States, financial institutions are subject to capital requirements and regulatory reporting rules. For example, insurers are required to maintain a minimal amount of capital based on risk assessments determined by the risk-based capital (RBC) ratio (total adjusted capital/required risk-based capital). Required risk-based capital is determined by the NAIC according to an insurer's business and asset risk. The NAIC classifies asset risk into six risk categories. Before the 2008 crisis, the risk categories for fixed-income securities like ABS are determined by credit ratings. Securities rated AAA, AA, or A are classified as Category 1; securities rated BBB, BB, B, and CCC are classified as Category 2, 3, 4, and 5, respectively; securities rated CC or below are classified as Category 6. The riskier an investment is, the higher risk factor it is assigned, and hence, the more capital is required.⁴

Insurers are also governed by different regulatory reporting rules: MTM and HCA. According to the NAIC, P&C insurers are required to employ the MTM rule for investments below Category 2 (i.e., below rating BBB), while life insurers can use the HCA rule to report securities unless they are in Category 6 (i.e., in default).⁵

Ellul et al. (2015) show that these different regulations distinctly affect insurers' trading strategies on ABS. Specifically, during the massive downgrade of ABS, HCA-governed life insurers continue to hold the downgraded ABS at historical costs, avoiding recognizing losses from price decline. To make up for the lowered RBC ratio (due to increased risk-weighted assets in the denominator of the ratio), they sell corporate and government bonds with the largest unrealized gains. Doing so allows them to realize gains and boost statutory capital (the numerator of the RBC ratio).

In contrast, MTM-governed P&C insurers have to immediately recognize the value erosions of troubled ABS. The resulting losses lower their capital and the capital adequacy ratio. To fulfill capital requirements, they fire-sell troubled

⁴ After the crisis, although the NAIC continues to rely on credit ratings for other securities, it adopts a new methodology—which relies on the expected loss instead of ratings—to determine the risk categories of residential MBS and commercial MBS.

⁵ If the securities held by life insurers are in Category 6 (i.e., rating CC or below) and are effectively in default, then they need to report the lower of the HCA and MTM values for their holdings. See the NAIC Note to *Schedule D—Part I on Long-term Bonds Owned December 31 of Current Year*.

ABS—despite the lack of market liquidity and severely depressed ABS prices amid massive downgrades. Doing so allows them to swap risky ABS for cash or other safer securities, reducing risk-weighted assets (the denominator of the RBC ratio) and relieving the capital requirements pressure.

Therefore, MTM—when combined with capital requirements—forces P&C insurers to realize ABS losses at the trough of market value and give up the option for subsequent price recovery. The anticipated potential losses for P&C insurers can in turn enforce the two conditions necessary for effective risk management (Ellul and Yerramilli 2013). First, risk managers may have stronger motives to perform thorough due diligence, which enables them to identify indications of ABS risk beyond apparent risk metrics. Specifically, risk managers can probe the ABS registration statements and evaluate the underlying collateral of the securitization pool, or infer the extent of ABS risk by examining the disclosure quality of deal prospectus, particularly the frequency of uncertain text usage in the prospectus (Liang et al. 2017).⁶ Risk managers may also scrutinize the stock selling of executives from the ABS underwriting banks, given that these executives are found to have exploited their private information on not-yet-reported nonperforming securitization income or securitization-related recourse risks to capture short-term profit ahead of the crisis (e.g., Bebchuk, Cohen, and Spamann 2010; Ryan, Tucker, and Zhou 2015; Cziraki 2018; Kolasinski and Yang 2018).⁷

Second, the anticipated ABS losses under MTM and capital requirements may curb executive incentives to invest excessively in ABS in the pursuit of profit, thereby promoting the recognition of ABS risk identified by risk managers (Calomiris 2008; Duhigg 2008; Coval, Jurek, and Stafford 2009; Gennailoli, Shleifer, and Vishny 2012; Erel, Nadauld, and Stulz 2014).⁸

On the other hand, under HCA, insurers' reported income and regulatory capital become less sensitive to the market value changes of security holdings and therefore are largely shielded from potential value declines in ABS. Such a shield affords risk managers some latitude to become lax in performing costly due diligence and enables executives with high-powered compensation to aggressively build up ABS exposures for their high yields, given that potential losses will not be promptly reflected in firms' reported income or executive compensation.

⁶ Information on ABS fundamentals is disclosed under the Regulation AB of the SEC since 2005. The regulation also requires an issuer to provide periodic disclosure on the changes in the underlying risk of the ABS.

⁷ Nevertheless, we acknowledge that adequately assessing the underlying ABS risk may be challenging, and risk managers' ability to successfully do so may be uncertain. This uncertainty works against us finding results in our examination of the "due diligence channel," which we discuss in the following.

⁸ Moreover, the MTM rule also provides executives and the board with more timely and transparent information regarding the market value of ABS, further strengthening executives' risk recognition. Such timely information is important because evidence shows that the managers of some large financial institutions seem to be unaware of their companies' excessive exposure to ABS, and therefore refuse to take the warnings of risk officers seriously (Dash and Creswell 2008).

Taken together, MTM, when combined with capital requirements, engenders more effective risk controls than the case of HCA and capital requirements in ABS investment.

Hypothesis 1. Risk management is more effective in constraining ABS exposure for P&C insurers than for life insurers.

P&C and life insurers may differ in firm characteristics and business natures other than the financial regulations they face. To the extent that these differences may lead to different risk preferences or prudence between the two types of insurers, they may also affect insurers' ABS investment strategies. To ensure that our results are not confounded by such differences, we explore the variations in market value recognition within the life insurance sector. In the United States, insurance regulations are implemented at the state level. State insurance commissioners have discretions when applying NAIC-model rules to insurers domiciled in their states. This discretion in turn generates variations in the recognition of market value. We therefore classify life insurers into two groups: those domiciled in states with a high level of market value recognition (analogous to the MTM-governed P&C insurers), and those in states with a low level of market value recognition (see Section 3.3). We hypothesize that under capital requirements, risk management of life insurers with a higher level of market value recognition is likely more effective in constraining ABS exposure compared with life insurers with a lower level of market value recognition. Because this analysis is performed within life insurers, it mutes confounding differences arising from business natures or firm characteristics between P&C and life insurers. Therefore, it allows us to better identify the effects of financial regulations on risk management.

Hypothesis 2. Risk management is more effective in constraining ABS exposure in life insurers domiciled in high-MTM states than in life insurers domiciled in low-MTM states.

We supplement the above hypotheses by considering the channels through which financial regulations affect the functioning of risk management. We consider two non-mutually exclusive channels. First, if MTM combined with capital requirements engenders more effective risk management by promoting greater due diligence of risk managers to probe the underlying ABS risk, then for insurers under MTM, risk management's function in constraining ABS investment should be stronger when such due diligence is most needed, that is, when an insurer holds ABS with a complex structure. In contrast, this incremental effect should be weaker among insurers governed by HCA. We term this channel the "due diligence" channel. Second, if MTM and capital requirements make risk management more effective because such financial regulations help curb executive incentives to neglect identified ABS risk and build up excessive ABS investment, then for insurers under MTM, risk management should be more effective in constraining ABS exposure when such incentives are

more prevalent, that is, when executive compensation grants more risk-taking incentives. Again, this incremental effect should be weaker among insurers governed by HCA. We term this channel the “risk-taking incentive” channel.

2. Data, Sample, and Summary Statistics

2.1 ABS investment

Our primary sample consists of U.S. public insurers obtained from the Compustat database (i.e., companies with SIC codes between 6311 and 6399). We focus on public insurers because, as we discuss in detail in Section 2.2, the information needed for measuring risk management strength is hand-collected from company filings. In our main analyses, we focus on how financial regulations affect the functioning of risk management during the period of 2005 to 2007. This is the period when the securitization market builds up considerable (yet lurking) credit risk (see Figure 1 and Table 1) and, hence, when risk management function is most germane and crucial. This period is also used in Ellul et al. (2014) to examine insurers’ asset allocations before the crisis. In Section 4.1, we extend our analyses back to 1998, and confirm our findings there.

We collect insurers’ ABS investment information from several sources, including Schedule D of the NAIC databases, Bloomberg, and S&P’s Ratings IQuery. Schedule D reports information on insurers’ securities investment. We merge Compustat insurers with Schedule D using company names. In cases where this merge is not successful, we match them using the names of an insurer’s subsidiaries, as reported in Exhibit 21 of 10-Ks.⁹ We next match securities held by insurers with Bloomberg data using CUSIPs to identify the type of each security. In certain cases where Bloomberg provides ambiguous information on security types, we refer to S&P’s Ratings IQuery. This step allows us to identify, and quantify, an insurer’s ABS investment in a given year. Following Erel, Nadauld, and Stulz (2014), ABS includes MBS, CDO, and other assets-backed securities. We focus on non-government-sponsored ABS because government-sponsored ABS are backed by government guarantees.

S&P’s Ratings IQuery records rating information for ABS rated by S&P. For each asset-backed security, we collect its historical rating assignment, rating changes, and the date of a rating action. This information enables us to investigate whether risk management can look beyond apparent risk assessments (e.g., ratings) and uncover the underlying risk of ABS reflected in ex post rating changes. In additional analyses, we also examine the reputation of ABS underwriters (Griffin, Lowery, and Saretto 2014) and the default status of ABS as alternative ABS risk measures. Lastly, we obtain insurers’ financial information from NAIC InfoPro to construct control variables.

⁹ This approach is based on the fact that a parent company often dictates its subsidiaries’ risk management policy (Eckles, Hoyt, and Miller 2014).

We consider insurers as ABS investors and study how financial regulations interact with risk management to affect their ABS investments. For this reason, we drop insurers that also serve as ABS underwriters to eliminate potential confounding effects resulting from the dual roles. In our sample, insurers rarely underwrite ABS. Based on the ABS underwriter information from S&P's ratings IQuery, we only observe two life insurers (affiliated with SunAmerica) that have served as ABS underwriters.

In Figure 2, we plot the nominal amount of ABS investment by all U.S. insurers (panel A), and the proportion of ABS investment to assets (panel B) from 1998 to 2007, separately for P&C and life insurers. These plots show that insurers are prominent ABS investors. Moreover, while P&C insurers reduce their ABS holdings before the crisis between 2005 and 2006 (panel B), life insurers continue to maintain ABS investment in the same period. These distinct patterns may reflect differential functions of risk management in ABS investment under different financial regulations. We test this hypothesis in our analyses.

2.2 Measuring the strength of risk management function

Following Ellul and Yerramilli (2013), we measure risk management strength by constructing a risk management index. The risk management index is constructed from five variables. The first four variables measure the presence and importance of the Chief Risk Officer (CRO). They include whether a CRO responsible for enterprise risk management is present (*CRO present*), whether the CRO is an executive officer (*CRO is executive*), whether the CRO is among the five highest paid executives (*CRO in top 5*), and the CRO's power in the organization captured by his or her relative compensation to the CEO (*CRO centrality*).

We hand-collect information on these variables from both proxy statements and 10-K filings. To determine whether there is an individual responsible for enterprise risk management, in each year we search SEC filings using keywords such as "risk," "enterprise," "erm," "cro," and "chief risk." When a keyword is found, we read the surrounding text. A few points are worth noting. First, the person in charge of risk management may not always carry the title of "chief risk officer," and this is especially true for small insurers. We identify the presence of a CRO based on a broader set of titles related to risk management, for example, "Senior Vice President, Finance, Risk and Investments," "Senior Vice President, Asset Liability and Risk Management," and "Vice President, Corporate Actuarial and Risk Management."¹⁰

Second, the compensation of a risk officer—which we use for the component *CRO centrality*—is not always disclosed in the executive compensation table in proxy statements. In this case, we follow Ellul and Yerramilli (2013) and

¹⁰ When there is no evidence suggesting the presence of an individual in charge of risk management, we do not assume a firm's chief financial officer (CFO) automatically takes this responsibility, unless we find supporting evidence.

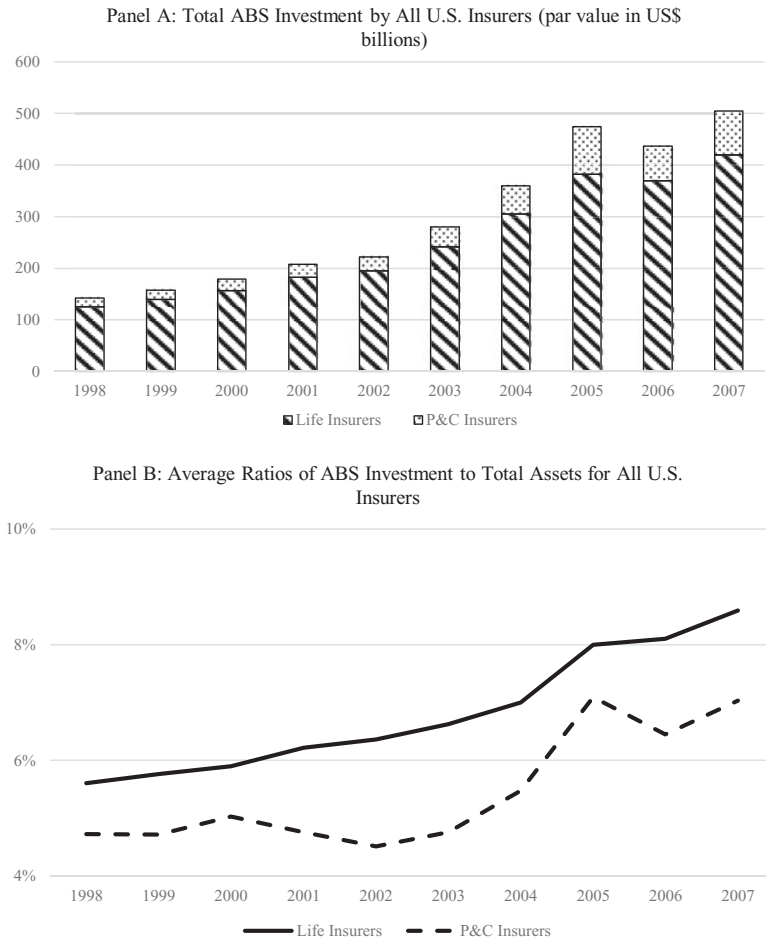


Figure 2
ABS investment by U.S. insurance companies

Panel A shows the aggregate par value of ABS investment by all U.S. life insurers and property and casualty (P&C) insurers, respectively, at each year end from 1998 to 2007. Panel B shows the average ratio of insurers' year-end par value of ABS investment to year-beginning total assets from 1998 to 2007.

calculate *CRO centrality* as the ratio of the compensation of the fifth highest-paid executive to the CEO's compensation minus 1%. This approach assumes that the (nondisclosed) CRO's compensation is just below the fifth highest-paid executive. In common with Ellul and Yerramilli (2013), we exclude the risk officer's stock and option awards from compensation when calculating *CRO centrality*. This treatment ensures that the compensation data we use to construct the risk management index (*RMI*) are consistent over 1998–2007, given that values of option grants are not systematically reported in earlier years.

Third, the *RMI* measure in Ellul and Yerramilli (2013) also considers the quality of risk oversight provided by the board of directors. They use the risk management committee's experience and the frequency of their meetings in BHCs to capture this dimension. However, these two pieces of information are typically not available for our sample insurers. For this reason, we use an alternative variable, *CRO report to board*, which is an indicator that equals 1 if the CRO reports to the board of directors, and 0 otherwise. A firm is considered to have stronger and more independent risk oversight if the CRO reports to the board instead of the CEO. This item is also used in Ellul and Yerramilli (2013) in robustness checks. Detailed definitions of these risk management variables are provided in the Appendix.

The five variables (i.e., *CRO present*, *CRO is executive*, *CRO in top 5*, *CRO centrality*, and *CRO report to board*) collectively capture the spirit of the recent regulatory efforts in strengthening financial institutions' internal risk management. For example, the 2010 Dodd-Frank Act, one of the most extensive financial regulatory acts since the 1930s, requires financial institutions to implement enterprise-wide risk management and to form dedicated risk committees that include qualified individuals to independently oversee the enterprise-wide risk management practices (see the Dodd-Frank Act, Section 165). We follow Ellul and Yerramilli (2013) to calculate the first principal component of these five variables in each year. The output is an annual *RMI*. A higher value of *RMI* indicates stronger risk management.

Table 2 presents summary statistics for the five components and the *RMI*. There are 2,407 insurer-year observations with available *RMI* in our sample over 2005–2007; 74% are P&C insurers. Forty-seven percent of the insurer-years have a CRO responsible for enterprise-wide risk management. The CRO has a rank at the executive level in about 25% of the insurer-years and is among the top five highest paid executives in about 5% of the sample. Based on *CRO centrality*, a CRO has average annual compensation that is approximately 17% of the CEO's compensation, and a CRO reports to the board 30% of the time. The *RMI* has a mean of 0.68. When we examine P&C and life insurers separately, we see that the average *RMI* is 0.73 for life insurers and 0.66 for P&C insurers. This difference, as well as the differences in *RMI* components, is not statistically significant.

Columns (4), (7), and (10) of Table 2 display the correlations between *RMI* and its five components for the full sample, P&C and life insurers, respectively. As expected, *RMI* is positively and significantly correlated with all five risk management components.

2.3 Corporate governance and other firm characteristics

Erkens, Hung, and Matos (2012) show that corporate governance affects financial institutions' risk taking during the crisis. Brav et al. (2008) find that activist institutional investors such as hedge funds play an important role in affecting corporate governance. We therefore include institutional ownership

Table 2
Summary statistics

	N (1)	Mean (2)	Std. dev. (3)	Corr with <i>RMI</i> (4)	N (5)	Mean (6)	Corr with <i>RMI</i> (7)	N (8)	Mean (9)	Corr with <i>RMI</i> (10)	<i>p</i> -value of diff. (11)
CRO present (0/1)	2,407	0.469	0.499	0.909***	1,775	0.455	0.905***	632	0.509	0.928***	0.257
CRO is executive (0/1)	2,407	0.250	0.433	0.779***	1,775	0.257	0.806***	632	0.229	0.697***	0.538
CRO in top 5 (0/1)	2,407	0.048	0.213	0.513***	1,775	0.053	0.548***	632	0.033	0.383***	0.431
CRO centrality	2,407	0.169	0.215	0.792***	1,775	0.162	0.783***	632	0.189	0.828***	0.446
CRO report to board (0/1)	2,407	0.300	0.459	0.757***	1,775	0.295	0.782***	632	0.316	0.686***	0.304
RMI	2,407	0.677	1.962	-	1,775	0.657	-	632	0.734	-	0.736
P&C insurers (0/1)	2,407	0.737	0.440	-0.017	1,775	1.000	-	632	0.000	-	-
Firm size	2,407	19.220	2.274	0.067***	1,775	18.787	-0.001	632	20.436	0.227***	0.000
RBC ratio	2,291	2.828	1.668	0.117***	1,662	2.994	0.171***	629	2.388	-0.050	0.000
Stock (%)	2,386	7.264	13.438	-0.023	1,758	7.590	-0.043*	628	6.350	0.031	0.181
Reinsurance	2,338	0.452	0.387	0.107***	1,722	0.539	0.128***	616	0.212	0.129***	0.000
HHI	2,170	0.402	0.371	-0.092***	1,586	0.430	-0.098***	584	0.327	-0.057	0.000
Firm age	2,341	3.393	0.841	0.052**	1,709	3.298	0.051**	632	3.652	0.047	0.000
Institutional ownership	2,400	0.653	0.249	0.304***	1,768	0.650	0.336***	632	0.660	0.210***	0.560
Independent director	2,400	0.738	0.144	0.200***	1,768	0.736	0.181***	632	0.743	0.253***	0.485
Female director	2,400	0.113	0.089	0.235***	1,768	0.108	0.254***	632	0.127	0.177***	0.233
ABS investment	2,407	4.585	7.311	-0.142***	1,775	3.876	-0.201***	632	6.577	0.001	0.000
Substantial downgrade	1,417	0.428	0.495	-0.015	918	0.332	-0.141***	499	0.603	0.112**	0.000
Rating downgrade notches	1,417	5.710	5.611	-0.042	918	4.811	-0.125***	499	7.364	0.036	0.000

This table reports the summary statistics of key variables in the period of 2005–2007 for all sample insurers, and for P&C insurers and life insurers, respectively. See Appendix for detailed variable definitions. Columns (4), (7), and (10) report the Pearson correlation between the risk management index (*RMI*) and each variable. Column (11) reports the *p*-value of the differences in the means of each variable between P&C and life insurers. *, **, and *** denote statistical significance at the 0.10, 0.05, and 0.01 levels (two-tailed), respectively.

and board independence in our analysis. We obtain a firm's institutional ownership (*Institutional ownership*) at the end of each fiscal year from the Thomson Financial Institutional Investor Ownership database. We calculate the proportion of independent directors (*Independent director*) to capture board independence. Furthermore, female directors are often more conservative and better monitors than male directors (e.g., Adams and Ferreira 2009). We therefore generate *Female director* to measure the proportion of female directors on the board.

We also generate a set of firm-level controls. First, as a firm's capital adequacy and hedging can affect risk taking (Bonaime, Hankins, and Harford 2014), we control for a firm's risk-based capital ratio (*RBC ratio*) and the proportion of reinsurance business—a proxy for liability risk hedging. Insurers often purchase reinsurance to limit underwriting liabilities and expand their risk-taking capacity (Mayers and Smith 1990). We follow Mayers and Smith (1990) and Adams, Hardwick, and Zou (2008) to calculate the ratio of reinsurance premiums ceded to the sum of the direct premiums written and reinsurance premiums assumed.¹¹ Second, we include the Herfindahl index of an insurer's collected premiums across different states as a measure of underwriting risk diversification. More diversified underwriting may reflect prudence or allow greater risk taking in investment. Lastly, we control for several standard firm characteristics, including firm size, age, and stock allocations. Detailed variable definitions are presented in the Appendix, and the summary statistics are reported in Table 2.

3. Main Results

3.1 Financial regulations and the role of risk management in ABS investment: P&C vs. life insurers

In this section, we test whether financial regulations affect the efficacy of risk management in constraining ABS investment (Hypothesis 1). We gauge the effectiveness of risk management from two aspects. First, following Ellul and Yerramilli (2013) and Ellul et al. (2014), we test whether strong risk management leads to a lower level of ABS investment. This approach is based on the intuition that ABS are generally considered risky due to the complex structure and the information asymmetry between ABS issuers and investors (Riddiough 1997; DeMarzo and Duffie 1999; Ghent, Torous, and Valkanov 2019). Second, we examine whether strong risk management helps uncover the underlying quality of ABS investments.

¹¹ Ideally, we would like to measure reinsurance by the amount of liability coverage taken up by reinsurance. However, such information is not available from public sources or NAIC regulatory databases. We therefore follow the literature and use the premium information to capture reinsurance, based on the assumption that the amount of premium is proportionate to liability coverage.

3.1.1 Level of ABS investment. We start by examining the relation between risk management and the level of ABS investment under different financial regulations. We estimate the following baseline ordinary least squares (OLS) regression in the period leading up to the crisis, that is, from 2005 to 2007. As we discussed in Section 1.1, this is a period when the securitization market builds up considerable lurking risk, and when risk management function is most crucial.

$$ABS\ investment_{i,t} = \alpha + \beta_1 RMI_{i,2004} + \beta_2 firm\ characteristics_{i,t-1} + \beta_3 corporate\ governance_{i,t-1} + fixed\ effects + \varepsilon_{i,t}, \quad (1)$$

where i indexes firm and t indexes time. Following Erel, Nadauld, and Stulz (2014), *ABS investments* is defined as the year-end total par value of the ABS held by an insurer scaled by its year-beginning total assets. The variable of interest is an insurer's risk management strength, *RMI*. We estimate Equation (1) separately for P&C and life insurers to identify risk management functions under different financial regulations. Based on Hypothesis 1, we expect β_1 to be significantly negative for P&C insurers, indicating that strong risk management lowers the level of ABS investment; this effect should be weaker for life insurers.

In the spirit of Ellul and Yerramilli (2013), we measure *RMI* at the end of 2004 (one year before the start of our sample period). Doing so allows us to examine whether a firm's risk management that is in place affects its ABS investment decisions during the period leading up to the crisis. All other firm-specific variables enter regressions with a one-year lag. We winsorize all variables at 1% of both tails to mitigate the effects of outliers.

Table 3, panel A, reports the results. Columns (1) to (3) pertain to P&C insurers, and Columns (4) to (6) concern life insurers. We start with a parsimonious model without controls, and then add in firm characteristics and corporate governance controls. We include year fixed effects to account for potential time trends in insurers' ABS investments. We cluster standard errors at the firm and year levels to account for possible correlations between years and across firms.

In Column (3) with both firm characteristic and corporate governance controls, the significant and negative coefficient of *RMI* indicates that stronger risk controls are associated with smaller ABS exposure for P&C insurers in the precrisis period. On the other hand, the insignificant coefficient of *RMI* in Column (6) suggests that the effect of risk management on constraining ABS investment is weaker for life insurers that are subject to HCA. The difference in the coefficients of *RMI* between Columns (3) and (6) is statistically significant at the 1% level, as reported at the bottom of the table. This difference is also economically sizable. Based on Column (3), a one-standard-deviation increase in *RMI* (1.962) is associated with a 1.42% drop in P&C insurers' ABS investment (relative to total assets), representing about 37% of the sample average for P&C insurers. Yet for life insurers, this effect is considerably weaker. These results suggest that financial regulations play an important role in affecting the functioning of risk management in ABS investment.

Table 3
The effect of financial regulations on the functioning of risk management

Panel A: Level of ABS investment

	P&C			Life	
	(1)	(2)	(3)	(4)	(6)
RMI	-0.664*** (-7.531)	-0.658*** (-6.972)	-0.723*** (-5.298)	0.149 (0.719)	-0.115 (-0.474)
Firm size		1.443*** (6.266)	1.433*** (7.411)		0.855*** (3.735)
RBC ratio		0.263 (0.941)	0.158 (0.551)		0.018 (0.046)
Stock (%)		-0.118*** (-8.055)	-0.104*** (-6.693)		-0.092*** (-4.818)
Reinsurance		-0.472 (-0.401)	-0.462 (-0.385)		-1.990 (-1.640)
HHI		-0.224 (-0.261)	0.253 (0.278)		0.329 (0.243)
Firm age		-0.971*** (-2.924)	-0.860*** (-2.731)		-0.811 (-0.806)
Institutional ownership			6.926*** (3.717)		2.988** (2.175)
Independent director			-4.379*** (-3.370)		-9.187*** (-4.539)
Female director			-6.921*** (-3.739)		-16.155*** (-3.294)
Year FE	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.043	0.132	0.188	0.003	0.162
N	1,775	1,454	1,447	632	578
<i>p-value of diff. in RMI coeff.</i>					(3)-(6) 0.000

Table 3
(Continued)

Panel B: Quality of ABS investment

	Substantial downgrade			Life			P&C			Downgrade notches			Life	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)		
RMI	-0.117*** (-3.496)	-0.173*** (-3.398)	-0.128*** (-4.117)	0.147*** (3.515)	-0.003 (-0.060)	-0.004 (-0.053)	-0.442*** (-2.845)	-0.449*** (-3.161)	-0.174** (-2.222)	0.260* (1.726)	-0.127 (-0.696)	0.025 (0.114)		
Firm size		0.313*** (5.438)	0.415*** (5.267)		0.239*** (3.499)	0.056*** (3.498)		0.564*** (2.501)	0.929*** (3.854)		0.079 (0.628)	0.196 (1.498)		
RBC ratio		0.059 (0.720)	-0.010 (-1.03)		0.391*** (3.100)	0.402*** (3.272)		-0.035 (-0.097)	-0.133 (-0.348)		0.927** (1.996)	0.795* (1.657)		
Stock (%)		0.006 (0.913)	-0.004 (-0.600)		0.011 (1.602)	0.009 (1.355)		0.057** (1.998)	0.009 (0.468)		0.083*** (2.978)	0.071*** (2.593)		
Reinsurance		-0.026 (-0.144)	0.179 (0.989)		-0.006 (-0.017)	0.049 (0.133)		0.101 (0.109)	0.706 (0.832)		1.655* (1.919)	1.746** (2.150)		
HHI		0.319** (2.246)	0.511*** (3.649)		-0.044 (-0.198)	0.101 (0.405)		1.185** (1.985)	1.565*** (3.065)		-1.272* (-1.897)	-0.784 (-1.177)		
Firm age		-0.039 (-0.435)	-0.006 (-0.062)		-0.035 (-0.303)	0.003 (0.028)		-0.238 (-0.814)	-0.213 (-0.734)		-0.037 (-0.110)	-0.021 (-0.065)		
Institutional ownership			1.076*** (4.752)			0.554* (1.718)			1.659* (1.665)			1.025 (1.098)		
Independent director			-1.878*** (-3.660)			-0.284 (-0.387)			-9.592*** (-4.209)			-4.009** (-2.060)		
Female director			-3.911*** (-7.942)			-1.508 (-1.590)			-10.459*** (-4.267)			-5.369*** (-3.236)		
Beginning rating FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes		
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes		
Pseudo/adjusted R ²	0.089	0.265	0.332	0.065	0.290	0.300	0.071	0.243	0.325	0.084	0.268	0.294		
N	896	802	795	486	460	460	896	802	795	486	460	460		
														(9) - (12)
														0.239

This table tests the effect of financial regulations on the functioning of risk management in constraining the level of ABS investment and in uncovering the underlying quality of ABS. Panel A reports OLS results from regressing the level of ABS investment on risk management strength (*RMI*) for the sample of NAIC insurers in the period of 2005–2007. The dependent variable is *ABS investment*, defined as the ratio (measured in percentage) of the total par value of ABS held by an insurer at the year end to the insurer's year-beginning total assets. *RMI* is measured in 2004. All other explanatory variables are measured at the beginning of a year. Panel B reports results from regressing the quality of ABS investment on *RMI* for the sample of NAIC insurers that hold a positive amount of ABS in the period of 2005–2007. Columns (1)–(6) present the probit estimation results where the dependent variable is *Substantial downgrades* (*0/1*), a dummy variable indicating whether any ABS held by an insurer is downgraded from AAA to a speculative grade as of 2009. Columns (7)–(12) present OLS results where the dependent variable is *Downgrade notches*, the par-value weighted average of notches by which an ABS held by an insurer is subsequently downgraded. Detailed variable definitions are described in the Appendix. *Year FE* are indicators for years. *Beginning rating FE* are dummies for rating categories of the par-value weighted average beginning ratings of ABS held by an insurer. *t*-statistics (in parentheses) are based on robust standard errors clustered at the firm and year levels. *p*-values of the differences in the regression coefficients of *RMI* are reported at the bottom of the table. *, **, and *** denote statistical significance at the 0.10, 0.05, and 0.01 levels (two-tailed), respectively.

As for other control variables, larger insurers have greater exposure to ABS. Insurers that hold more stocks (*Stock%*) invest less in ABS. To the extent that larger stock holdings indicate an insurer's stronger preference for liquid assets, this observation is consistent with Gorton (2009), who views ABS as assets with relatively low liquidity. Second, most corporate governance controls significantly affect ABS investment. This observation is different from prior studies that often find a trivial effect of corporate governance on banks' risk taking in the crisis (e.g., Fahlenbrach and Stulz 2011; Beltratti and Stulz 2012; Erkens, Hung, and Matos 2012). Specifically, insurers with higher institutional ownership (who are more likely to face pressures to pursue short-term profits) have larger ABS investment. Firms with more independent directors and a higher proportion of female directors on the board invest less in ABS, in line with the monitoring role of female directors found in Adams and Ferreira (2009).

3.1.2 Quality of ABS investment. Next, we examine how financial regulations affect the relation between risk management and the quality of insurers' ABS investment. To capture the quality of ABS, we start by considering the underlying risk reflected in ex post credit rating changes. Credit ratings are widely used by both investors and regulators to assess ABS risk. However, it is not uncommon to see that an asset-backed security rated as AAA ex ante is substantially downgraded to a speculative grade ex post (Table 1). In this case, the ex ante favorable ratings have masked the underlying risk of ABS. If risk management can look beyond the apparent risk metrics and help insurers uncover the underlying quality of ABS investment, then we expect insurers with stronger risk management to be less exposed to ABS that carry favorable ratings ex ante but get substantially downgraded ex post.¹²

For each year, we track the rating changes of an ABS, and determine whether the ABS is downgraded as of 2009. If in a given year, an insurer holds an ABS that is downgraded from AAA to a speculative-grade rating (i.e., below BBB), then we let the *Substantial downgrade* dummy equal 1 (and 0 otherwise). More generally, we also calculate by how many notches an ABS held by an insurer in a given year is downgraded as of 2009.¹³ We then define the par-value weighted average of these notches for the insurer as *Downgrade notches*. This measure gauges the overall decline in ABS credit quality. A higher value of *Substantial downgrade* or *Downgrade notches* indicates lower ABS quality.

¹² We would like to note that our measure of ABS quality is not without limitations. Specifically, poor performance ex post does not necessarily mean ineffective risk management ex ante (Stulz 2008). In our context, it is possible that P&C and life insurers may have different risk management objectives, and certain insurers do not have the objective to improve the ex post rating performance. In this case, a failure to avoid rating downgrades would not necessarily indicate a failure to fulfill these insurers' risk management goals. To account for potential different objectives of risk management, in Section 3.3 we perform analyses using only life insurers, and such analyses are less susceptible to this concern. We confirm our findings there. We thank an anonymous reviewer for pointing out this possibility.

¹³ We do so by assigning a numerical value to S&P ratings (1 = AAA, 2 = AA+, 3 = AA, 4 = AA- ..., 22 = D).

Table 3, panel B, presents the results using the two variables. We perform the analyses in the sample where insurers have positive ABS holdings, and therefore when ABS rating information is available. Columns (1) to (6) pertain to *Substantial downgrade*, and Columns (7) to (12) concern *Downgrade notches*. In Columns (1) to (6), we estimate the following probit model:

$$\text{Prob}(\text{Substantial downgrade}_{i,t}) = \Phi(\alpha + \beta_1 \text{RMI}_{i,2004} + \beta_2 \text{firm characteristics}_{i,t-1} + \beta_3 \text{corporate governance}_{i,t-1} + \text{fixed effects} + \varepsilon_{i,t}). \quad (2)$$

where i indexes firm and t indexes time. In Columns (7) to (12), we estimate the following OLS model:

$$\text{Downgrade notches}_{i,t} = \alpha + \beta_1 \text{RMI}_{i,2004} + \beta_2 \text{firm characteristics}_{i,t-1} + \beta_3 \text{corporate governance}_{i,t-1} + \text{fixed effects} + \varepsilon_{i,t}. \quad (3)$$

As before, we estimate Equations (2) and (3) separately for P&C and life insurers to identify any differential effects of risk management on uncovering the underlying ABS quality. In both equations, we control for firm characteristics, corporate governance, and year fixed effects. We also include fixed effects for ABS beginning ratings to capture nonlinear effects of ratings on downgrade probabilities (e.g., a AAA-rated ABS may have a different downgrade probability than a CCC-rated ABS). Specifically, for each year, we calculate the par-value weighted average rating for ABS held by an insurer and generate five indicators corresponding to rating categories of AAA, AA, A, BBB, and the rest, respectively.

In Equations (2) and (3), we expect the coefficient of *RMI*, β_1 , to be significantly negative for P&C insurers, which would indicate that risk management leads them to invest in ABS with lower lurking risk and hence higher quality. This coefficient, however, should be weaker for life insurers. This is indeed what we find. P&C insurers with stronger risk management are less likely to invest in ABS that bear a favorable rating but are later substantially downgraded (Columns (1) to (3)). Their ABS investments also have smaller downgrade magnitudes (Columns (7) to (9)). Based on Column (3), the marginal effects of *RMI* (evaluated at the mean of *RMI*) on decreasing the probability of substantial downgrade for P&C insurers is 4.2%. It is economically sizable compared with the 33% average downgrade probability of the sample P&C insurers. Based on Column (9), a one-standard-deviation increase in *RMI* lowers the downgrade magnitude of P&C insurers' ABS investment by 0.34 notches, and the sample average for P&C insurers is 4.81 notches. Consistent with the results from panel A of Table 3, this role of risk management is considerably weaker for life insurers. Using both *Substantial downgrade* and *Downgrade*

notches measures, the coefficients of *RMI* for life insurers in Columns (6) and (12) are insignificant.¹⁴

Overall, these results suggest that strong risk management not only lowers insurers' levels of ABS investment but also helps uncover the lurking risk of most problematic ABS. The effectiveness of risk control, however, hinges on financial regulations.

3.2 Propensity score matching

P&C and life insurers may differ along dimensions other than the financial regulations they are subject to. For example, one may argue that P&C insurers face higher uncertainty than life insurers due to less predictable liabilities, shorter duration of funding, larger stock holdings, or a higher geographical concentration in insurance underwriting (see Table 2). Such higher uncertainty may result in greater prudence and lower risk tolerance, making them less inclined to hold risky ABS given the same level of risk management strength as life insurers. In this case, it may appear that risk management is more effective in constraining P&C insurers' ABS holdings than for life insurers, whereas this relation merely reflects differences in the underlying uncertainty related to P&C and life insurers' characteristics. Alternatively, one may argue these different characteristics can result in different risk profiles and asymmetric information that insurers deal with, which in turn contribute to the different reporting rules that P&C and life insurers face. In this case, our findings may be an artifact of the endogenous financial regulations.

As a first step to address these concerns, we use a propensity score matching method. The idea is to obtain a set of balanced covariates between P&C and life insurers so that their different characteristics are trimmed into comparable groups. Under the assumption that confounding factors are reflected in observable firm characteristics, the matching method creates a setting in which these confounding factors are likely to affect different insurers' ABS holdings in a similar way, and hence, it helps us isolate the effect of financial regulations on the functioning of risk management.

For each year, we match a life insurer with a P&C insurer based on a host of firm characteristics—including *RMI*, *Firm size*, *RBC ratio*, *Stock(%)*, *Reinsurance*, *HHI*, *Firm age*, and credit ratings of an insurer's ABS holdings.¹⁵ We follow Rosenbaum and Rubin (1985) in implementing the matching. First, we run a logit model among sample P&C and life insurers by regressing a *Life* indicator (that equals 1 if an insurer is a life insurer, and 0 otherwise) on all matching dimensions in each year. We generate a propensity score—that is, the predicted probability of an insurer being a life insurer based on the logit model.

¹⁴ We note that although the difference in the coefficient of *RMI* between P&C and life insurers is reasonably large in magnitude between Column (9) and Column (12), it is not statistically significant. This is likely due to the large standard error of the *RMI* coefficient for life insurers.

¹⁵ We match life insurers with P&C insurers because there are fewer life insurers than P&C insurers.

Next, for each life insurer in a given year, we select a P&C insurer that has the closest propensity score (i.e., the nearest neighbor) as the matched P&C insurer. To further ensure that the matched two groups are comparable, we calculate the distance in the propensity score between a life insurer and its matched P&C insurer and drop any match whose distance in the propensity score is above the 99th percentile of the distance distribution.

We start by performing various diagnostic tests to verify that the matched P&C and life insurers share balanced covariates. In panel A of Table 4, we report the p -values testing the differences in the means of firm characteristics between the two groups. We see that all firm characteristics that have shown significant differences between the unmatched P&C and life insurers (see Column (11) of Table 2) now become statistically indistinguishable from each other. The economic magnitudes of remaining differences also become trivial. In Table IA1 panel A of the Online Appendix, we further report the standardized differences and variance ratios of these characteristics, before and after the matching separately. We show that after the matching, the standardized differences of the characteristics are close to zero and variance ratios are close to one, again suggesting that there are no systematic differences in these characteristics between the matched pairs.¹⁶

Next, we repeat our main analyses for the matched sample and test whether financial regulations affect the efficacy of risk management between the matched P&C and life insurers. Since the two groups of insurers already share balanced covariates, we start with a univariate test. That is, we compare whether P&C insurers' ABS investment is more responsive to stronger risk management than that of life insurers (Hypothesis 1). The responsiveness is measured by the Pearson correlation between ABS level (or ABS quality) and *RMI*. If risk management is more effective in constraining ABS exposure under the MTM and capital requirements, then we should see a more negative correlation between ABS level (or ABS substantial downgrades) and *RMI* for P&C insurers than for life insurers. For brevity, we only use *Substantial downgrade* to represent ABS quality.

Panel B of Table 4 presents this univariate test. The correlation between ABS level and *RMI* is -0.322 for P&C insurers, and it is 0.001 for life insurers. The difference of the two correlations is statistically significant at the 1% level (based on Fisher's [1915] z -transformation), as shown in Column (3). This observation is consistent with our hypothesis. We obtain a similar observation for the correlations between ABS quality and *RMI*.

In panels C and D, we repeat the same regression analyses as in Table 3 for matched P&C and life insurers. Panel C pertains to the level of ABS investment, and panel D concerns the quality of ABS investment. The results confirm our previous findings. The negative and significant coefficients of

¹⁶ Rubin (2001) advises that the absolute standardized differences below 0.25 and variance ratios within the range of (0.5, 2) indicate reasonably good matching.

RMI in Columns (1) and (2) of both panels indicate that for P&C insurers, strong risk management not only lowers the level of ABS investment, but also helps uncover the underlying risk of ABS. On the other hand, Columns (3) and (4) show that the effect of risk management is substantially weaker and insignificant at the 10% level among life insurers. In Table IA1, panel B, of the Online Appendix, we employ a stricter specification to compare the role of

Table 4
Propensity score matching

Panel A: Student *t*-tests of the differences of sample means between matched P&C and life insurers

	P&C	Life	<i>p</i> -value of difference
	(1)	(2)	(3) = (1) – (2)
RMI	0.440	0.581	0.709
Firm size	21.226	21.234	0.976
RBC ratio	2.170	2.156	0.863
Stock (%)	7.421	5.026	0.200
Reinsurance	0.206	0.191	0.527
HHI	0.224	0.271	0.202
Firm age	3.791	3.778	0.915
Beginning ratings	2.043 ($\approx$ AA+)	2.251 ($\approx$ AA+)	0.225

Panel B: Univariate tests of the responsiveness of ABS investment to risk management strength

	P&C	Life	<i>p</i> -value of difference
	(1)	(2)	(3) = (1) – (2)
Level of ABS investment	–0.322	0.001	0.000
Quality of ABS investment	–0.072	0.201	0.000

Panel C: Level of ABS investment

	P&C		Life	
	(1)	(2)	(3)	(4)
RMI	–2.311** (–2.126)	–1.884* (–1.749)	–0.082 (–0.279)	0.228 (0.751)
Firm size	3.319* (1.784)	3.321* (1.887)	0.343 (1.287)	0.568* (1.843)
RBC ratio	2.764 (1.125)	2.354 (1.164)	0.380 (0.616)	0.208 (0.393)
Stock (%)	–0.140 (–0.913)	–0.165 (–1.340)	–0.089*** (–2.930)	–0.112*** (–3.569)
Reinsurance	–4.272 (–1.195)	–3.903 (–1.356)	–1.541 (–0.987)	–1.063 (–0.782)
HHI	3.263 (0.843)	4.382 (1.407)	1.895 (1.100)	3.625** (2.041)
Firm age	2.006 (0.870)	1.684 (0.769)	–0.212 (–0.187)	0.047 (0.042)
Institutional ownership		13.389* (1.793)		3.921** (2.464)
Independent director		–2.485 (–0.669)		–6.546** (–2.000)
Female director		–24.735*** (–3.025)		–20.364*** (–2.952)
Year FE	Yes	Yes	Yes	Yes
Adjusted R^2	0.476	0.533	0.031	0.109
N	448	448	448	448
<i>p</i> -value of diff. in RMI coeff.				(2) – (4) 0.000

Table 4
(Continued)

Panel D: Quality of ABS investment

	P&C		Life	
	(1)	(2)	(3)	(4)
RMI	−0.184*** (−4.204)	−0.125*** (−4.499)	−0.008 (−0.166)	−0.007 (−0.126)
Firm size	0.389*** (3.496)	0.456*** (9.212)	0.225*** (3.214)	0.252*** (3.277)
RBC ratio	0.11 (0.389)	0.047 (0.167)	0.354** (2.136)	0.379** (2.468)
Stock (%)	0.030** (2.150)	0.022 (1.298)	0.007 (1.247)	0.006 (1.010)
Reinsurance	−0.714 (−0.990)	−0.519 (−0.819)	0.031 (0.092)	0.101 (0.282)
HHI	1.079 (1.382)	1.093 (1.172)	−0.083 (−0.384)	0.076 (0.321)
Firm age	0.381 (1.153)	0.267 (0.655)	−0.069 (−0.583)	−0.03 (−0.249)
Institutional ownership		1.301 (1.366)		0.604* (1.913)
Independent director		−2.234* (−1.832)		−0.378 (−0.531)
Female director		−2.717 (−1.461)		−1.427 (−1.484)
Beginning rating FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Pseudo R^2	0.41	0.452	0.275	0.286
N	448	448	448	448
				(2) − (4) 0.107

p-value of diff. in RMI coeff.

This table presents analyses of propensity score matching. In each year, a life insurer is matched with a P&C insurer, based on firm characteristics including *RMI*, *Firm size*, *RBC ratio*, *Stock (%)*, *Reinsurance*, *HHI*, *Firm age*, and credit ratings of insurers' ABS holdings. Propensity score matching follows the procedure of Rosenbaum and Rubin (1985) and is described in Section 3.2. Panel A presents diagnostic tests to verify that the matched P&C insurers and life insurers share balanced covariates. It reports the sample means (Columns (1) and (2)), as well as *p*-values testing the differences of sample means between P&C insurers and life insurers for each variable (Column (3)). Panel B reports univariate tests for the responsiveness of insurers' ABS investment to risk management strength (*RMI*). The responsiveness of insurers' ABS investment level is calculated as the Pearson correlation between the level of ABS investment and *RMI*. The responsiveness of insurers' ABS investment quality is calculated as the Pearson correlation between the quality of ABS investment (as represented by *Substantial downgrade*) and *RMI*. The level and quality of ABS investment is defined in the same way as in Table 3. The *p*-value testing the differences of the correlations, based on Fisher's (1915) *z*-transformation, is reported in Column (3). Panels C and D report a multivariate analysis, resembling panels A and B of Table 3, respectively. The dependent variables are *ABS investment (%)* and *Substantial downgrades*, as defined in Table 3. *RMI* is the risk management index measured in 2004, and all other independent variables are measured at the beginning of a given year. Detailed variable definitions are described in the Appendix. *Year FE* are indicators for years. *Beginning rating FE* are dummies for rating categories of the par-value weighted average beginning ratings of ABS held by an insurer. Robust standard errors clustered at the firm and year level are used in tests. *p*-values of the differences in the regression coefficients of *RMI* are reported at the bottom of the table. *, **, and *** denote statistical significance at the 0.10, 0.05, and 0.01 levels (two-tailed), respectively.

risk management on insurers' ABS investment within the matched P&C and life insurer pairs.¹⁷ We again confirm our findings.

¹⁷ Specifically, we pool the matched P&C and life insurers, and estimate the following model with the matched P&C/life insurer-pair fixed effects: $ABS\ level\ (or\ quality)_{i,t} =$

$$\begin{aligned} &\alpha + \beta_1 RMI_{i,2004} * PC\ insurer_i + \beta_2 RMI_{i,2004} + \beta_3 PC\ insurer_i \\ &+ \beta_4 firm\ characteristics_{i,t-1} + \beta_5 corporate\ governance_{i,t-1} + PC\ life\ pair\ fixed\ effects \\ &+ other\ fixed\ effects + \varepsilon_{i,t}. \end{aligned}$$

3.3 Baseline analyses within the life insurance sector

To further ensure that our findings are not confounded by different business natures between P&C and life insurers, we perform analyses within the life insurance sector and test Hypothesis 2. The U.S. insurance industry is largely regulated at the state level. State insurance commissioners have discretion when applying the NAIC-model rules to insurers domiciled in their states.¹⁸ This discretion generates variations in the extent of market value recognition and, therefore, determines how close a state's insurance rule is to the MTM rule for life insurers. Based on these variations, we classify life insurers into two groups: those domiciled in states with a high tendency of market value recognition (analogous to the MTM-governed P&C insurers), and those in states with a low tendency of market value recognition. We then compare the functioning of risk management between these two groups.

The analysis within the life insurance sector is important in helping us minimize differences that arise between P&C and life insurance companies (including different business models, risk attitudes, or risk management objectives) because such differences are arguably much less prominent among insurers operating in the same sector. Therefore, this analysis supplements the propensity score matching and absorbs factors that may not be fully captured by the observable firm characteristics we use in the matching. Moreover, because state insurance codes are established well before our sample period, and because insurers rarely change their states of domicile, variations in market value recognition in this setting are largely exogenous to insurers' ABS investment and risk management.¹⁹

Following Ellul et al. (2014), for each state, we calculate the average frequency of life insurers carrying bonds of speculative grades at the fair value over 2002–2004 (the period before our main period of analysis). If the frequency of a state is above the national median, then we classify this state as a high-MTM state; otherwise, the state belongs to the low-MTM category.²⁰

We repeat the analyses in Table 3 within life insurers and report the results in Columns (1) to (4) of Table 5. The results support Hypothesis 2. For life insurers in high-MTM states, strong risk management not only significantly lowers the level of ABS investment (Column (1)), but also improves the quality

¹⁸ The NAIC merely acts as a forum for creating model laws and regulations. Although NAIC models are widely adopted and provide some degree of uniformity among states, they do not bear the authority of law and have no effect unless they are adopted by a state.

¹⁹ The rarity of insurers' change in the states of domicile is also reported in Ellul et al. (2015, p. 2500).

²⁰ Note that this identification is based on state insurance commissioners' actual implementation of the market value recognition. Alternatively, one may also classify high-MTM states and low-MTM states by reading state-level insurance codes and inferring from these codes whether a state is likely to require market value recognitions for insurers domiciled in that state. However, Ellul et al. (2015, p. 2501) point out that as with any interpretation of rules, this approach may yield ambiguous inferences regarding the adoption of market value recognition in some states.

Table 5
Analyses within the life insurance sector

	Full sample				Propensity score matched sample			
	Level of ABS investment		Quality of ABS investment		Level of ABS investment		Quality of ABS investment	
	High MTM (1)	Low MTM (2)	High MTM (3)	Low MTM (4)	High MTM (5)	Low MTM (6)	High MTM (7)	Low MTM (8)
RMI	-0.695** (-2.369)	0.365 (1.213)	-0.323*** (-3.664)	0.070 (0.653)	-0.888*** (-3.608)	-0.011 (-0.011)	-0.310*** (-3.450)	0.397*** (4.079)
Firm size	0.591** (1.981)	1.206*** (3.510)	0.461* (1.742)	0.228*** (3.281)	0.022 (0.070)	0.851 (0.909)	0.457* (1.707)	0.321** (2.065)
RBC ratio	0.747 (0.993)	-0.345 (-0.571)	1.316** (2.040)	0.191 (1.303)	0.110 (0.184)	1.330 (0.735)	1.326* (1.951)	-0.043 (-0.128)
Stock (%)	-0.154** (-2.260)	-0.129*** (-4.406)	-0.015 (-1.146)	0.018** (2.105)	-0.190 (-1.158)	-0.022 (-0.361)	0.001 (0.095)	0.001 (0.075)
Reinsurance	-1.774 (-1.387)	-1.934 (-1.076)	-0.425 (-0.906)	0.235 (0.579)	-3.490** (-2.410)	-0.893 (-0.213)	-0.511 (-1.116)	1.286 (1.491)
HHI	-5.382*** (-3.683)	2.881 (1.612)	-0.699 (-1.133)	0.107 (0.407)	-7.890*** (-3.598)	0.508 (0.111)	-0.753 (-1.191)	0.051 (0.115)
Firm age	-1.553* (-1.663)	-0.801 (-0.584)	-0.444 (-0.813)	0.022 (0.162)	-2.195* (-1.917)	-2.914 (-0.874)	-0.521 (-0.911)	-0.521 (-1.102)
Institutional ownership	2.179 (0.826)	2.332 (1.504)	1.390** (1.981)	0.146 (0.339)	-0.224 (-0.059)	2.616 (0.863)	1.327* (1.901)	1.132 (0.840)
Independent director	6.209** (2.028)	-16.385*** (-3.745)	3.326** (2.482)	-1.184 (-0.902)	9.960*** (4.031)	-7.770 (-0.680)	3.190*** (2.752)	-5.832*** (-3.271)
Female director	-17.172*** (-2.809)	-14.133** (-2.169)	-1.418 (-0.583)	-0.268 (-0.159)	-15.293* (-1.900)	-25.402 (-1.371)	-1.009 (-0.423)	-0.197 (-0.059)
Beginning rating FE	No	No	Yes	Yes	No	No	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo/adjusted R^2	0.256 197	0.213 369	0.562 149	0.249 299	0.236 146	0.192 146	0.377 146	0.377 146
N		(1) - (2)	(3) - (4)	(3) - (4)		(5) - (6)	(7) - (8)	(7) - (8)
p -value of diff. in RMI coeff.		0.021	0.004	0.004		0.238		0.000

This table reports the regression analyses for the sample of NAIC life insurers over 2005–2007. We follow Ellul et al. (2014) and classify life insurers into two groups: the ones domiciled in a high mark-to-market (high-MTM) state and the ones in a low mark-to-market (low-MTM) state. Columns (1)–(4) use all sample life insurers. Columns (5)–(8) use a propensity-score-matched sample, where we match a life insurer in a high-MTM state to a life insurer in a low-MTM state in each year, based on *RMI*, *Firm size*, *RBC ratio*, *Stock (%)*, *Reinsurance*, *HHI*, *Firm age*, and credit ratings of ABS holdings. The dependent variables are *ABS investment (%)* and *Substantial downgrades* (0/1), as defined in Table 3. *RMI* is the risk management index measured in 2004, and all other independent variables are measured at the year-beginning. Detailed variable definitions are described in the Appendix. *Year FE* are indicators for years. *Beginning rating FE* are dummies for rating categories of the par-value weighted average beginning ratings of ABS held by an insurer. t -statistics (in parentheses) are based on robust standard errors clustered at the firm and year levels. p -values of the differences in the regression coefficients of *RMI* are reported at the bottom of the table. *, **, and *** denote statistical significance at the 0.10, 0.05, and 0.01 levels (two-tailed), respectively.

of ABS investment (Column (3)). In contrast, the effect of risk management is considerably weaker for life insurers in low-MTM states, as indicated in Columns (2) and (4).

3.4 Propensity score matching within the life insurance sector

The baseline within-life-insurers analyses help us mute differences between P&C and life insurers. However, it is still possible that certain differences exist in key characteristics of life insurers between the high-MTM and low-MTM states. To further minimize these remaining differences, we perform propensity score matching within life insurers. We match life insurers in high-MTM states with those in low-MTM states, following a similar matching procedure as described in Section 3.2.²¹ We first verify that the matched life insurers in high- and low-MTM states share balanced covariates (results untabulated). We then perform the main analyses based on the matched sample and report the results in Columns (5) to (8) of Table 5. We confirm our findings: the effect of risk management on constraining ABS exposure is only significant for life insurers in high-MTM states, but not for life insurers in low-MTM states.

The positive coefficient of *RMI* in Column (8) of Table 5 (i.e., for the matched life insurers domiciled in low-MTM states) deserves discussion. Note that our hypothesis predicts that for life insurers in high-MTM states, risk management plays a significant role in constraining ABS exposure; for life insurers in low-MTM states, such a role should be weaker. In the latter case, however, we do not have a strong prior on how risk management should be associated with ABS investment. The positive and significant coefficient of *RMI* we observe in Column (8), combined with the insignificant coefficient in Column (6), suggests that among matched life insurers in low-MTM states, stronger risk management is associated with a higher likelihood of ABS being substantially downgraded, though it does not change the level of ABS investments. Therefore, it seems that stronger risk controls afford such insurers to carry out more reaching-for-yield strategies (Becker and Ivashina 2015). However, this observation should be interpreted with caution because we do not find a similar result among the full sample of life insurers in low-MTM states (Columns (2) and (4) of Table 5), or among all life insurers (Table 3).

3.5 Discussion of the main analyses

Our findings from Sections 3.1 to 3.4 suggest that regulatory reporting rules, in combination with capital requirements, play an important role in affecting the functioning of risk management in ABS investment. Amid financial industry regulators' endeavors to improve risk management in financial institutions, these findings provide an important policy implication.

²¹ We match life insurers in high-MTM states with life insurers in low-MTM states because there are fewer life insurers in high-MTM states in our sample.

Merely installing risk management systems into financial institutions may not be sufficient. Regulators should consider the value of external financial regulations, specifically the prompt recognition of investment losses enforced by MTM and capital requirements. These considerations are salient given that institutions are typically not incentivized to voluntarily recognize investment losses in a prompt manner and therefore do not internalize the benefits of such external regulations (FCIC 2011).

To supplement our main findings, we also consider the channels through which financial regulations affect the functioning of risk management. We present detailed discussions and analyses in Section IA.2, Tables IA2 and IA3, of the Online Appendix. There we show suggestive evidence that financial regulations affect risk management functions through two channels: promoting risk managers' efforts in uncovering ABS risk and curbing executives' incentives to take excessive risk.

Our findings highlight the interaction between financial institutions' internal risk management and external financial regulations. The thrust of these findings lies in that MTM, when combined with capital requirements, promotes risk recognition, which leads to more effective risk management. Therefore, our findings can be generalized to financial institutions' other types of investments (e.g., corporate bonds) or to other time periods (e.g., the reemerging collateralized loan obligation [CLO] market after the crisis)—as long as the promptness of risk recognitions of these investments depends on reporting rules.²² On the other hand, our results have limited implications for insurers' liabilities derived from the underwriting business.²³ This is because unlike investments, insurers' liabilities do not typically carry an observed market value; therefore, reporting rules—MTM and HCA—have limited effect on the recognition of such liabilities.²⁴

3.6 Does risk management affect insurers' investment performance and tail risk?

Our results so far show that stronger risk management not only reduces insurers' level of ABS investment, but also leads them to invest in ABS that have lower lurking risk under MTM and capital requirements. A natural question is how risk management affects insurers' performance. Specifically, does strong risk management at the onset of the financial crisis mitigate the adverse effect of ABS investment on insurers' overall investment performance and tail risk

²² According to Standard & Poor's, CLOs made an astonishing comeback after fading into obscurity during the 2008 financial crisis. In 2017, the CLO issuance topped \$118 billion in the United States, close to the record level (see <http://www.leveragedloan.com/surpassing-expectations-2017-us-clo-issuance-nears-record-levels/>).

²³ For example, the liability risk due to changing longevity for life insurers, or cyber risk for P&C insurers.

²⁴ Indeed, as shown in Table 2, stronger risk management is associated with more hedging of underwriting risk via insurance, and this association is significant in both MTM-governed P&C insurers and HCA-governed life insurers.

during the crisis? If so, does this effect vary with different financial regulations that insurers face?

We start with examining insurers' performance of ABS investment, and then examine insurers' overall investment performance and tail risk. Following Ellul et al. (2014), we measure an insurer's ABS performance as the sum of ABS realized gains on sales or maturity and unrealized gains, minus other-than-temporary-impairments (OTTI), divided by year-beginning total assets (*ABS performance*). A positive (negative) value of *ABS performance* indicates a gain (loss). We then estimate the regression model below:

$$ABS\ performance_{i,t} = \alpha + \beta_1 RMI_{i,2007} + \beta_2 firm\ characteristics_{i,t-1} + \beta_3 corporate\ governance_{i,t-1} + fixed\ effects + \varepsilon_{i,t}, \quad (4)$$

where i indexes firm and t indexes time. This model is estimated during the financial crisis period between 2008 and 2009. The variable of interest, *RMI*, is measured as of 2007. It therefore tells us whether an insurer's existing risk management (*RMI*) prior to the financial crisis affects its ABS investment performance in the crisis. Given that strong risk management can constrain P&C insurers' ABS exposure, especially to the risky ones as we previously showed, we expect the coefficient, β_1 , to be positive for P&C insurers. On the other hand, we expect this effect to be weaker for life insurers.

Table 6 presents the results. Column (1) reports the results for P&C insurers. Column (4) reports the results for life insurers. As expected, strong risk management significantly increases P&C insurers' ABS investment performance, but this effect of *RMI* is considerably weaker for life insurers. It suggests that the value implications of risk management vary with different financial regulations.

We next test whether strong risk management at the onset of the financial crisis mitigates the adverse effect of ABS investment on insurers' overall investment performance and tail risk during the crisis. We measure total investment return as the total annual investment income obtained from the income statement divided by year-beginning total assets. We follow Ellul and Yerramilli (2013) and define tail risk as the negative of the average return of an insurer's stock during the 5% worst return days in a year. We estimate an augmented specification of Equation (4), where we add the interaction between *ABS* and *RMI* ($ABS_{i,2007} * RMI_{i,2007}$) in the estimation. This specification helps us understand whether insurers' existing ABS investment at the onset of the crisis (i.e., as of 2007) has a negative (positive) effect on their overall investment performances (tail risk) in the crisis, and more importantly, whether this effect can be mitigated by strong risk management, as indicated by the coefficient of the interaction term, $ABS * RMI$.

This is indeed what we find in Columns (2), (3), (5), and (6) of Table 6. Columns (2) and (5) pertain to total investment return; Columns (3) and (6) concern tail risk. In Columns (3) and (6), we include additional controls that

Table 6

The value implications of risk management

	P&C			Life		
	ABS performance (%) (1)	Total investment return (%) (2)	Tail risk (%) (3)	ABS performance (%) (4)	Total investment return (%) (5)	Tail risk (%) (6)
RMI*ABS		1.547*** (8.537)	−3.609*** (−4.370)		−1.578 (−1.567)	2.921 (1.341)
ABS		−3.698*** (−15.531)	5.411*** (2.693)		−3.829*** (−3.125)	10.663*** (3.096)
RMI	0.042*** (4.230)	−0.129 (−0.979)	−0.010 (−0.046)	−0.002 (−0.127)	0.005 (0.057)	−0.051 (−0.141)
Firm size	−0.080** (−2.318)	−0.164*** (−2.961)	0.361*** (2.624)	−0.032* (−1.706)	−0.396*** (−4.537)	0.234* (1.700)
RBC ratio	0.040*** (2.594)	0.115** (1.973)	0.381** (2.223)	0.030 (0.596)	−0.353 (−1.176)	1.563*** (4.499)
Stock (%)	0.001 (0.421)	−0.073 (−0.729)	−0.032 (−1.533)	0.000 (0.028)	−0.083 (−1.249)	0.088** (2.428)
Reinsurance	−0.329*** (−3.661)	−0.804 (−1.400)	1.012* (1.866)	0.018 (0.183)	0.607 (1.373)	−0.675* (−1.884)
HHI	0.044 (0.478)	0.115 (0.325)	−2.259* (−1.902)	−0.246*** (−3.501)	−0.038 (−0.019)	−1.277 (−1.094)
Firm age	0.150*** (2.676)	0.444*** (3.778)	−0.221 (−0.611)	−0.077 (−1.055)	0.534 (1.488)	−0.683** (−2.409)
Institutional ownership	−0.144 (−0.713)	1.654 (1.579)	−3.797 (−1.352)	−0.267 (−1.009)	1.398 (0.694)	0.521 (0.212)
Independent director	0.050 (0.182)	0.751 (1.022)	−8.679 (−1.469)	0.102 (0.140)	−2.323 (−0.857)	−8.232*** (−3.545)
Female director	0.635*** (2.871)	4.074*** (3.675)	−3.256 (−0.895)	0.246 (0.306)	2.881 (0.724)	17.899*** (4.653)
CEO vega		0.238***			0.273**	
CEO delta			(3.458) −0.000 (−0.262)			(2.559) −0.000*** (−3.251)
Past returns			−6.902 (−1.602)			−6.827*** (−8.692)
Beginning rating FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.249	0.348	0.532	0.158	0.234	0.572
N	442	442	414	217	217	185
				(1) – (4)	(2) – (5)	(3) – (6)
<i>p</i> -value of diff. in coeff.				0.075 (RMI)	0.093 (RMI*ABS)	0.006 (RMI*ABS)

This table examines the value implications of risk management under different financial regulations. All dependent variables are measured in the crisis period 2008–2009. In Columns (1) and (4), the dependent variable is *ABS performance* (%), defined as (realized gains on sales or maturity + unrealized gains – other-than-temporary-impairments [OTTI] each year) / year-beginning total assets. The variable of interest is the risk management index, *RMI*, measured in 2007. In Columns (2) and (5), the dependent variable is *Total investment return* (%), defined as the total investment income obtained from the income statement in a year divided by year-beginning total assets. In Columns (3) and (6), the dependent variable is *Tail risk* (%), defined as the negative of the average return on a stock during the 5% worst return days in a year. The variable of interest in Columns (2)–(3) and (5)–(6) is the interaction between an insurer's ABS holding at the end of 2007 (*ABS*) and *RMI* (measured in 2007), which tests whether strong risk management at the onset of the financial crisis mitigates the adverse effect of ABS investment on insurers' overall investment performance and tail risk in the crisis. *Year FE* are indicators for years. *Beginning rating FE* are dummies for rating categories of the par-value weighted beginning average ratings of ABS held by an insurer. *t*-statistics (in parentheses) are based on robust standard errors clustered at the firm and year levels. *p*-values of the differences in the regression coefficients are reported at the bottom of the table. *, **, and *** denote statistical significance at the 0.10, 0.05, and 0.01 levels (two-tailed), respectively.

measure CEO compensation characteristics (*CEO delta* and *CEO vega*) and past year stock returns (*Past returns*) following Ellul and Yerramilli (2013). We see that as expected, the coefficient of *ABS* is significantly negative in the model of total investment return (suggesting that ABS exposure lowers insurers' overall investment return), whereas the coefficient of *ABS**RMI** is significantly positive at the 1% level among P&C insurers. Similarly, the coefficient of *ABS* is significantly positive in the model of tail risk (suggesting that ABS exposure increases insurers' overall tail risk), whereas the coefficient of *ABS**RMI** is significantly negative among P&C insurers. Also as expected, the interaction terms of *ABS**RMI** are not significant for life insurers in either model. Consistent with the notion that a higher CEO vega promotes risk taking, the coefficient of *CEO vega* is positive and significant for both P&C and life insurers.

Overall, our results suggest that strong risk management can mitigate the adverse effect of ABS investment on insurers' overall investment performance and tail risk, and this effect depends on external financial regulations. Given that existing studies document that ABS were largely overvalued before the crisis, and the ex ante yields of these securities did not sufficiently compensate for the underlying lurking risk (e.g., Piskorski, Seru, and Witkin 2015; Griffin and Maturana 2016; Ghent, Torous, and Valkanov 2019; Kolasinski and Yang 2018), it is unlikely that insurers' exposure to ABS is optimal. Hence, by constraining such an exposure, risk management appears to be value enhancing for P&C insurers.

4. Additional Analyses

4.1 Analyses in an extended sample period and on the unloading of ABS investment

Our main analyses focus on the sample period between 2005 and 2007. As we discussed in Section 1, this period is when the securitization market built up considerable lurking risk and, hence, when risk management's role in uncovering such risk is most crucial. In this section, we extend the sample period to ten years back from the onset of the crisis in 2007 (i.e., 1998–2007) and check the robustness of our results. To construct variables for the extended sample, we follow the same procedures as described in Section 2. That is, we manually collect information for *RMI* and corporate governance variables from SEC filings, and we obtain insurers' ABS investment and other characteristics from InfoPro. We repeat our main analyses and report the results in Table 7.

Columns (1) to (4) pertain to the level of ABS investment, and Columns (5) to (8) concern the quality of ABS investment. We estimate the same regressions as in Equations (1) and (2), except that here we use $RMI_{i,t-1}$ (i.e., lagged *RMI*) instead of *RMI* measured in 2004. We do so to better suit the longer panel data in the extended sample.

We start with analyses on ABS levels. Consistent with previous findings, the negative coefficient of *RMI* in Column (1) indicates that stronger risk

Table 7
Analyses over the extended sample between 1998 and 2007

	Level of ABS investment			Quality of ABS investment				
	P&C		Life	P&C		Life		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
RMI	-0.482*** (-6.011)	-0.524*** (-5.492)	-0.076 (-0.597)	-0.063 (-0.352)	-0.098*** (-3.709)	-0.302*** (-7.757)	-0.012 (-0.358)	-0.076 (-1.258)
Firm size	0.814*** (4.394)	0.679*** (2.778)	0.806*** (5.072)	0.977*** (2.587)	0.325*** (4.834)	0.688*** (3.445)	0.259*** (4.787)	0.774*** (3.599)
RBC ratio	-0.103 (-0.716)	-0.108 (-0.788)	-0.269 (-1.057)	-0.219 (-0.702)	-0.040 (-0.521)	-0.072 (-0.433)	0.352*** (2.591)	0.335** (1.826)
Stock (%)	-0.063*** (-5.506)	-0.058*** (-5.632)	-0.086*** (-5.114)	-0.070*** (-3.011)	-0.001 (-0.267)	-0.010 (-0.758)	0.002 (0.251)	-0.014 (-0.943)
Reinsurance	-0.765 (-1.613)	-0.415 (-0.808)	-0.932 (-1.378)	-1.153 (-1.547)	0.031 (0.153)	0.034 (0.121)	-0.195 (-0.662)	-0.069 (-0.158)
HHI	-0.166 (-0.403)	-0.585 (-1.305)	0.673 (0.594)	1.250 (1.048)	0.407*** (2.348)	0.588* (1.823)	-0.233 (-1.014)	-0.409 (-1.073)
Firm age	-0.537*** (-2.557)	-0.551*** (-2.490)	-0.273 (-0.534)	0.186 (0.314)	-0.065 (-0.707)	-0.065 (-0.342)	-0.138 (-1.168)	-0.502*** (-2.063)
Institutional ownership	2.915*** (2.721)	3.141*** (2.798)	0.579 (0.454)	0.093 (0.063)	0.664*** (2.113)	0.248 (0.316)	0.260 (0.731)	0.122 (0.197)
Independent director	1.561 (1.263)	1.707*** (2.010)	-2.674** (-2.089)	-1.834 (-1.527)	-0.822 (-1.205)	0.652 (0.650)	0.146 (0.370)	-0.443 (-0.935)
Female director	-4.471** (-2.268)	-4.634* (-1.901)	-13.143*** (-3.701)	-14.148*** (-3.593)	-3.236*** (-5.452)	-5.411*** (-3.076)	-0.808 (-1.078)	-1.085 (-0.862)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Size decile FE	No	Yes	No	Yes	No	Yes	No	Yes
Beginning rating FE	No	No	No	No	Yes	Yes	Yes	Yes
Pseudo/adjusted R ²	0.147	0.158	0.128	0.176	0.366	0.447	0.312	0.431
N	4,589	3,761	1,859	1,337	1,460	609	954	490
			(1) - (3)	(2) - (4)			(5) - (7)	(6) - (8)
			0.000	0.000			0.033	0.003
p-value of diff in RMI coefficients								

This table tests the effect of financial regulations on the functioning of risk management in constraining the level of ABS investment and in uncovering the underlying quality of ABS for NAIC insurers over 1998–2007. Columns (1)–(4) report results from regressing the level of ABS investment on risk management strength (*RMI*), where the dependent variable, *ABS investment* (%), is defined as the ratio of the total par value of ABS held by an insurer at the year end to year-beginning total assets. Columns (5)–(8) report the results from regressing the quality of ABS investment on *RMI*, where the dependent variable is *Substantial downgrades (0/1)*, a dummy variable indicating whether any ABS held by an insurer is downgraded from AAA to a speculative grade. All explanatory variables are measured at the beginning of a year. Detailed variable definitions are described in the Appendix. *Beginning rating FE* are dummies for rating categories of the par-value weighted average beginning ratings of ABS held by an insurer. *Size decile FE* are based on firms' total assets as of 1998. Columns (5)–(8) are based on probit estimations, and other columns are based on OLS estimations. *t*-statistics (in parentheses) are based on robust standard errors clustered at the firm and year levels. *p*-values of the differences in the regression coefficients of *RMI* are reported at the bottom of the table. *, **, and *** denote statistical significance at the 0.10, 0.05, and 0.01 levels (two-tailed), respectively.

management leads to a lower ABS exposure for P&C insurers. This effect, albeit negative, is significantly weaker for life insurers, as shown in Column (3). The difference in the coefficients of *RMI* is statistically significant, as indicated by the *p*-value reported at the bottom of the table.

In Columns (2) and (4), we include size-decile fixed effects, in the spirit of Ellul and Yerramilli (2013). That is, we divide our sample insurers into ten decile groups based on their total assets in 1998 (the beginning year of the extended sample period) and include dummy variables indicating each size group in the estimation. To the extent that size is a significant determinant of institutions' enterprise-wide risk management strength (Ellul and Yerramilli 2013), and to the extent that our risk management measure may reflect certain time-invariant factors (such as insurers' risk preferences), the period-beginning size-decile fixed effects can capture such time-invariant factors and help partially absorb the confounding effect from these factors. Because these size-decile fixed effects are defined in 1998, we require that an issuer exist as of 1998 for the sample of analyses on ABS level, and additionally that an issuer has a positive holding of ABS as of 1998 for the sample of analyses on ABS quality (proxied by the substantial rating downgrade of ABS holdings). The results in Columns (2) and (4) are qualitatively similar to those in Columns (1) and (3). In Columns (5) through (8), we examine ABS quality and obtain similar interpretations.

To further control for time-invariant risk preferences of institutions, we include institution fixed effects in the estimations of both the level and quality of ABS investment. In unreported results, we observe that the coefficients of *RMI* among P&C insurers become statistically insignificant, which in turn makes the difference in *RMI* coefficients between P&C and life insurers insignificant. It is possible that the inclusion of institution fixed effects drives out variations of *RMI* in a given firm (including the variations of *RMI* arising from different financial regulations). In Section 4.2, we revisit this point by utilizing a difference-in-differences setting that allows us to identify significant variations in *RMI* surrounding the 9/11 terrorist attacks. There we confirm that our findings are not likely to simply reflect time-invariant factors such as institutions' risk preferences.

In the extended sample, we also examine the unloading of insurers' ABS investment at the onset of the crisis. Specifically, our results in Section 3.1 show that before the financial crisis, strong risk management helps constrain P&C insurers' ABS exposure under MTM and capital regulations. A related question is whether strong risk management enables insurers to unload their ABS positions before the crisis. On one hand, strong risk controls can help insurers act on indications of lurking risk in ABS investment, especially under MTM that provides more transparent investment performance information to top management and the board. On the other hand, it is unclear whether risk controls enable insurers to accurately time the market and reduce ABS holdings before the financial crisis unfolds. In the Online Appendix, Table IA4, we explore this issue as an empirical question. We show that for P&C insurers,

strong risk management significantly increases the likelihood that insurers unload their ABS investment, especially ABS with the highest lurking risk. However, the effect is weaker for life insurers, in line with our main findings.

4.2 Difference-in-differences analyses

4.2.1 Baseline analyses. In Sections 3.2 to 3.4, we discussed the concern that certain omitted variables—such as uncertainties related to insurers' business or risk attitude—may make P&C insurers less inclined to hold ABS than life insurers, even with the same level of risk management strength as life insurers. In this case, it may appear that risk management is more effective in reducing the ABS exposure for P&C insurers than for life insurers. To address this issue, we have used a propensity score matching (based on observable firm characteristics) and within-life-insurers analysis (which mutes differences between P&C and life insurers) to alleviate this concern. In this section, we supplement these analyses by exploiting changes in firms' risk management strength following an exogenous event and examine whether our findings withstand a difference-in-differences (DID) setting, which is more resilient to unobserved confounding factors.

The DID analysis relies on the setting of the terrorist attacks in September 2001. The 9/11 attacks heightened the importance of risk management for the insurance industry. Following 9/11, insurers strengthened their risk controls to different extents in response to this catastrophe. These changes in turn generate variations in the strength of risk management. On the other hand, the 9/11 event was virtually unexpected, and it occurred for reasons unrelated to the risk management or ABS investment of any particular firm. Hence, the variations in risk management strength generated by the 9/11 attacks can be reasonably viewed as insurers' different responses to an exogenous event.²⁵

Based on this intuition, we create a DID setting: we compare how insurers' ABS investment changes before and after the 9/11 event (the first difference) between insurers that experience a large increase in risk management strength (*RMI*) following 9/11 versus the ones that experience a smaller increase (the second difference). While the first difference may simply capture a common time trend in insurers' ABS investment that coincides with the occurrence of the 9/11 attacks, the second difference nets it out and allows us to isolate whether risk management has a direct effect on insurers' ABS decisions. If such an effect exists, we expect to see that insurers with a large increase in *RMI* following the 9/11 attacks experience a significant decrease in ABS exposure, relative to the other group.

We identify the two groups of insurers based on the extents of their *RMI* increase following the 9/11 attacks, separately for P&C and life insurers. The treatment group consists of insurers that experience a large increase in *RMI*, and

²⁵ In a similar vein, Ellul and Yerramilli (2013) construct an instrument for BHCs' risk management based on the 1998 Russian debt crisis.

the control group consists of insurers that experience a smaller increase in risk management strength. Specifically, for each insurer, we calculate average *RMI* during 1998–2001 (i.e., up to three years before the year when the 9/11 attacks occurred) and during 2002–2004 (i.e., up to three years after the 9/11 attacks). We then take the difference of the insurer's average *RMI* during the two periods to obtain the change in *RMI* surrounding 9/11. If an insurer's change in *RMI* surrounding 9/11 is above the 75th percentile of the *RMI* change distribution, then this insurer is categorized as a treatment firm. Otherwise it is categorized as a control firm.

We start with the DID tests for P&C insurers. We first plot how insurers' *RMI* changes surrounding 9/11 for the treatment and control groups. This analysis enables us to validate an important condition for the DID analysis: the parallel trend. Panel A of Figure 3 reports P&C insurers' *RMI* from 1998 to 2007. We only keep insurers that appear in both the pre-event period (i.e. 1998–2001) and post-event period (2002–2007) for the DID analyses. Panel A shows that before 9/11, the *RMI* of both treatment and control group stays relatively flat. The changes in *RMI* during the pre-event period for the two groups are statistically indistinguishable from each other (with a *p*-value of 0.22).

This parallel trend, however, alters following the 9/11 attacks. While the *RMI* of the control group continues to stay flat throughout the period until 2007, the *RMI* of the treatment group increases sharply after 9/11. It peaks in 2003, and decreases slightly over 2004–2007, presumably reflecting insurers' relatively lax risk controls amid the ABS market boom. Nevertheless, the *RMI* of the treatment group stays at a relatively high level compared with that of the control group throughout the post-event period. As indicated in Figure 3, the average *RMI* for the control group is slightly lowered, by 0.554, from the pre- to post-event period, whereas the average *RMI* for the treatment group increases by 1.971. Such a difference is both statistically and economically significant, suggesting that the dispersion in *RMI* generated by the 9/11 attacks appears to significantly alter the preexisting heterogeneity in *RMI* between the treatment and control groups.²⁶

We next perform the main DID analyses and examine whether the increase in *RMI* for the treatment group leads to a change in ABS decisions. We start by presenting graphical analyses in panel B of Figure 3, where we plot the level of ABS investment for the treatment and control groups between 1998 and 2007. We again observe that the ABS levels for the two groups exhibit a similar parallel trend up to 9/11, and diverge following 9/11. While the ABS investment of the control group substantially increases, especially during the ABS market expansion (starting around 2004), the treatment group does not follow suit. Their ABS investment level first trends downward, along with rising *RMI*, and stays at a relatively low level throughout the post-event period. This

²⁶ In untabulated analyses, we find that the *RMI* increase for the treatment group is driven by simultaneous increases in *CRO present*, *CRO report to board*, *CRO is executive*, and *CRO centrality* among the five components of *RMI*.

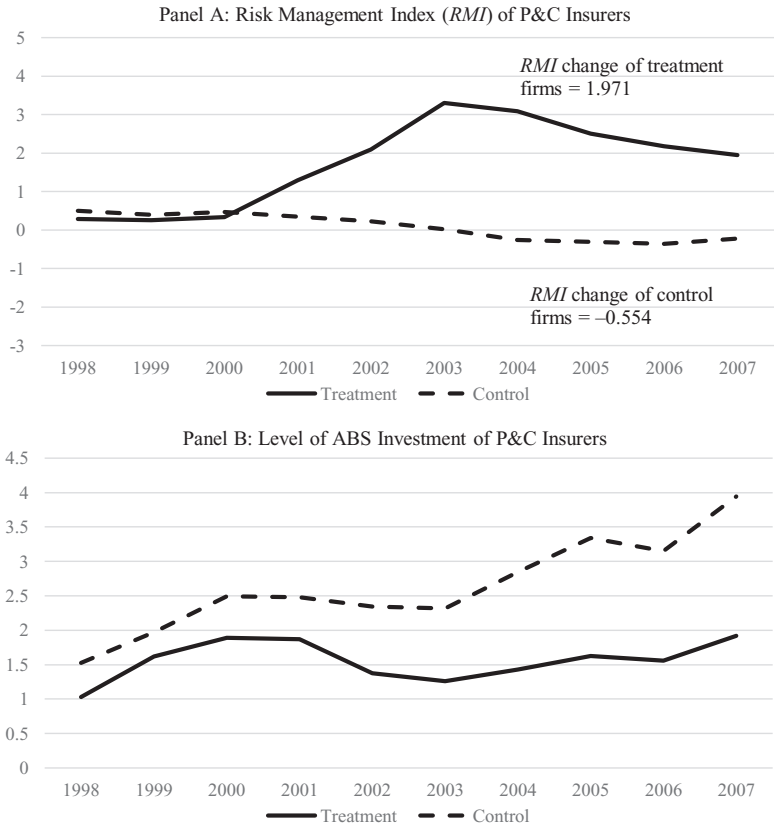


Figure 3

P&C insurers' *RMI* and level of ABS investment for the treatment and control groups

This figure plots P&C insurers' level of risk management index (panel A) and ABS investment (panel B) between 1998 and 2007, for the treatment group (represented by the solid line) and control group (represented by the dash line), respectively. The treatment and control groups are defined based on the distribution of P&C insurers' *RMI* changes surrounding the terrorist attacks in September 2001. For each sample P&C insurer, we calculate the average *RMI* during 1998–2001 (i.e., up to three years before 9/11) and during 2002–2004 (i.e., up to three years after 9/11), and the difference in the average *RMI* during the two periods is the insurer's change in *RMI* surrounding 9/11. If an insurer's change in *RMI* is above the 75th percentile of the *RMI* change distribution, the insurer is categorized as a treatment firm. Otherwise it is categorized as a control firm. In panel A, the numbers next to the lines indicate the average change in *RMI* from the pre-9/11 period (i.e., 1998–2001) to the post-9/11 period (i.e., 2002–2007), for the treatment and control groups, respectively. In panel B, the level of ABS investment is defined as the ratio (measured in percentage) of the total par value of ABS held by an insurer at the year end to the insurer's year-beginning total assets.

observation provides preliminary evidence that heightened risk management in the treatment group constrains the exposure to ABS, as compared with the control group.

Next, we formalize the graphical analyses by running the following DID regression:

$$\begin{aligned}
 \text{ABS investment}_{i,t} = & \alpha + \beta_1 \text{Treatment} * \text{Post} + \beta_2 \text{Treatment} \\
 & + \beta_3 \text{firm characteristics}_{i,t-1} \\
 & + \beta_4 \text{corporate governance}_{i,t-1} + \text{Year dummies} + \varepsilon_{i,t}. \quad (5)
 \end{aligned}$$

The dependent variable is the level of ABS investment, defined the same way as in Table 3. *Treatment* equals 1 if an insurer is in the treatment group, and 0 if it is in the control group. *Post* equals 1 if an observation is from the period after 9/11 (2002–2007), and it equals 0 if it is from the pre-event period (1998–2001). We include year dummies in the model, which subsume the dummy variable *Post*. The DID estimate is β_1 . A negative coefficient would indicate that relative to the control group, the ABS investment of the treatment group decreases significantly from the pre-event period to the post-event period, due to strengthened risk management.

Columns (1) to (3) of Table 8, panel A, present the results. In Column (1), we present a parsimonious specification that does not include firm characteristics or governance variables. The coefficient of *Treatment*Post* is -0.901 . It is statistically significant at the 1% level. In Column (2), we include firm characteristics, and in Column (3), we further add corporate governance variables as controls. In addition to the governance variables themselves, we also include the interaction of these variables with *Post* to account for the potential differential effects of governance controls on ABS investment before and after the passage of the Sarbanes-Oxley Act of 2002 (SOX), which roughly coincided with 9/11. In all specifications, we see a negative coefficient of the DID estimator *Treatment*Post*.

Next, we repeat the above analyses in life insurers. Panel A of Figure 4 first shows that similar to that of P&C insurers in Figure 3, the *RMI* pattern between the treatment and control groups also closely tracks each other up to 9/11, and then diverge following that. The *RMI* increase for the treatment group amounts to 1.576, while it decreases by 0.348 for the control group. This difference is significant at the 1% level, although the economic magnitude of the difference is smaller than that for P&C insurers. This is expected given that the impact of 9/11 is more moderate for life than for P&C insurers (Nelson 2015).

Despite the significant increase in *RMI*, the treatment group does not see a decline in ABS investment following 9/11—which is in contrast to the findings of P&C insurers. This result can be seen in both a graphical analysis (panel B of Figure 4) and a regression analysis (Columns (4) to (6) of Table 8, panel A). In particular, in Columns (4) to (6), we find that the coefficients of *Treatment*Post* are not significantly different from zero.²⁷

²⁷ The signs of these (insignificant) DID coefficients for life insurers are positive. It is consistent with the pattern in panel B of Figure 4: the ABS level of the treatment group is lower than the control group before 9/11, but quickly catches up after 9/11. As a result, the treatment group experiences a larger increase in ABS investment than the control group.

Table 8
Difference-in-differences analyses

Panel A: Full sample

	P&C			Life		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment*Post	−0.901*** (−3.292)	−0.963*** (−3.318)	−0.960** (−2.275)	0.693 (1.097)	0.643 (1.122)	1.046 (1.454)
Treatment	−0.516* (−1.753)	−0.475 (−1.459)	−1.112** (−2.391)	−0.805 (−1.290)	−1.109* (−1.692)	−0.954 (−1.259)
Firm size		0.764*** (4.602)	0.637*** (3.709)		0.466*** (3.364)	0.594*** (4.112)
RBC ratio		−0.101 (−0.751)	−0.104 (−0.772)		−0.443* (−1.828)	−0.467** (−2.071)
Stock (%)		−0.069*** (−6.030)	−0.050*** (−5.029)		−0.054*** (−2.868)	−0.063*** (−2.883)
Reinsurance		−0.391 (−0.833)	−0.570 (−1.203)		0.143 (0.178)	0.080 (0.104)
HHI		−0.825* (−1.675)	−0.857** (−2.065)		−0.056 (−0.050)	0.618 (0.526)
Firm age		−0.679*** (−2.867)	−0.534** (−2.519)		−0.169 (−0.411)	−0.164 (−0.408)
Institutional ownership			0.365 (0.439)			−0.239 (−0.152)
Independent director			3.812*** (4.788)			−1.862 (−1.343)
Female director			−1.202 (−0.513)			−5.099** (−2.569)
Institutional ownership*Post			4.570*** (3.073)			−0.113 (−0.076)
Independent director*Post			−2.079 (−1.405)			0.594 (0.327)
Female director*Post			−6.167** (−2.330)			−3.120 (−0.952)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.022	0.110	0.157	0.019	0.085	0.113
N	4,529	4,099	4,033	1,609	1,503	1,461

Panel B: Treatment firms with business exposure in New York and Washington, DC

	P&C			Life		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment*Post	−0.822** (−2.453)	−0.776** (−2.283)	−0.750* (−1.706)	0.370 (0.596)	0.491 (0.889)	0.950 (1.351)
Treatment	−0.613* (−1.952)	−1.125*** (−2.835)	−1.853*** (−3.901)	−0.638 (−1.011)	−1.148* (−1.749)	−1.019 (−1.342)
Firm size		0.788*** (4.551)	0.644*** (3.642)		0.485*** (3.304)	0.620*** (4.008)
RBC ratio		−0.148 (−1.044)	−0.152 (−1.113)		−0.480* (−1.796)	−0.504** (−2.029)
Stock (%)		−0.072*** (−5.932)	−0.050*** (−4.668)		−0.054*** (−2.835)	−0.062*** (−2.825)
Reinsurance		−0.406 (−0.802)	−0.613 (−1.201)		0.241 (0.285)	0.195 (0.241)
HHI		−1.104** (−2.025)	−1.131** (−2.493)		0.029 (0.025)	0.764 (0.623)
Firm age		−0.791*** (−3.138)	−0.636*** (−2.845)		−0.101 (−0.236)	−0.092 (−0.218)
Institutional ownership			0.546 (0.592)			−0.073 (−0.046)
Independent director			4.417*** (6.083)			−1.726 (−1.235)
Female director			−0.651 (−0.257)			−5.263*** (−2.638)

Table 8
(Continued)

	P&C			Life		
	(1)	(2)	(3)	(4)	(5)	(6)
Institutional ownership*Post			4.595*** (2.849)			-0.266 (-0.180)
Independent director*Post			-2.779* (-1.712)			0.405 (0.221)
Female director*Post			-6.968** (-2.508)			-3.258 (-0.962)
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.020	0.114	0.169	0.016	0.085	0.114
N	4,075	3,715	3,653	1,567	1,466	1,424

This table presents difference-in-differences analyses to examine whether the increase in insurers' risk management strength (*RMI*) following the 9/11 terrorist attacks leads them to lower the ABS exposure in the subsequent years. The sample period is from 1998 to 2007. *Treatment* is an indicator variable that equals 1 if an insurer belongs to the treatment group, and 0 if it belongs to the control group. The treatment and control groups are defined based on the distribution of insurers' *RMI* changes surrounding the 9/11 attacks. For each sample P&C (life) insurer, we calculate the average *RMI* during 1998–2001 (i.e., up to three years before the 9/11 attacks) and during 2002–2004 (i.e., up to three years after the 9/11 attacks), respectively, and then take the difference in average *RMI* during the two periods to obtain the insurer's *RMI* change surrounding the 9/11 attacks. If an insurer's *RMI* change is above the 75th percentile of the *RMI* change distribution of all sample P&C (life) insurers, the insurer is categorized as a treatment firm; otherwise it is a control firm. *Post* is an indicator variable that equals 1 if an observation is after the 9/11 attacks (from 2002 to 2007), and 0 if the observation is before the 9/11 attacks (from 1998 to 2001). The dependent variable is *ABS investment* (%), defined as the ratio of the total par value of ABS held by an insurer at the year end to its year-beginning total assets. Panel A consists of all sample insurers. In panel B, we drop insurers in the treatment group that do not have business exposures in New York and Washington, D.C. An insurer's business exposure to a state is determined based on the underwriting premiums collected by the insurer in that state as of 2000. In both panels, all other independent variables are measured at the beginning of a year. Detailed variable definitions are described in the Appendix. *Year FE* are indicators for years. *t*-statistics (in parentheses) are based on robust standard errors clustered at the firm and year levels. *, **, and *** denote statistical significance at the 0.10, 0.05, and 0.01 levels (two-tailed), respectively.

The different DID results between P&C and life insurers are important to us. If risk management simply captures confounding factors (such as risk attitudes) between the two types of insurers, and if our results are merely driven by these confounding factors, then we would have seen the treatment groups of P&C and life insurers both demonstrate a significant reduction in ABS investment following the large increase of *RMI* after 9/11 (relative to their control groups). But this is not what we find. It therefore further suggests that the different efficacy of risk management between P&C and life insurers is most likely attributed to financial regulations they face, rather than to confounding factors.²⁸

4.2.2 Insurers based in New York and Washington, D.C. To further sharpen the DID analyses, we consider insurers' business geography in relation to the

²⁸ Note that in the DID analyses, we only consider the level of insurers' ABS investment, but not the quality of ABS investment, as captured by substantial downgrades of ABS. This is because, as we discussed in Section 3.1.2, the measure of *Substantial downgrade* is available only for insurers with positive ABS holdings. During the pre-event period (before 2001), ABS issuance is rather limited, and insurers with positive ABS holdings are even more sparse. We require an insurer to appear in both the pre- and post-event periods for the DID analyses, and therefore we are left with a rather small sample for the tests of *Substantial downgrade*.

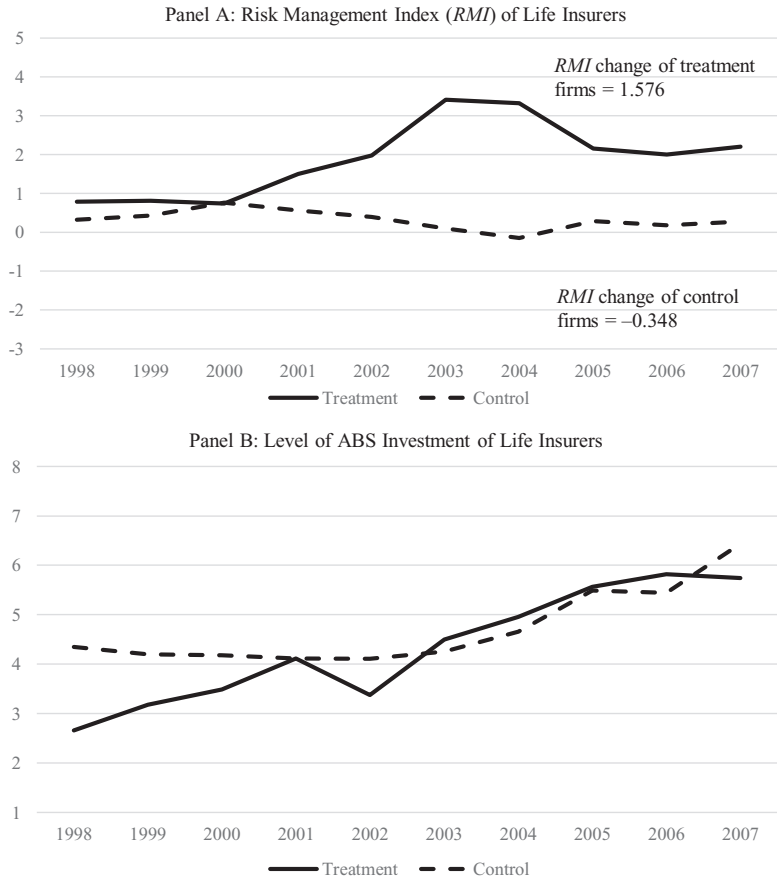


Figure 4
Life insurers' *RMI* and level of ABS investment for the treatment and control groups

This figure plots life insurers' level of risk management index (panel A) and ABS investment (panel B) between 1998 and 2007, for the treatment group (represented by the solid line) and control group (represented by the dash line), respectively. The treatment and control groups of life insurers are defined in the same way as in Figure 3. In panel A, the numbers next to the lines indicate the average change in *RMI* from the pre-9/11 period (i.e., 1998–2001) to the post-9/11 period (i.e., 2002–2007), for the treatment and control groups, respectively. In panel B, the level of ABS investment is defined as the ratio (measured in percentage) of the total par value of ABS held by an insurer at the year end to the insurer's year-beginning total assets.

9/11 attacks. The idea is as follows. Although it is reasonable to believe that the 9/11 event has heightened the importance of risk management for the whole insurance industry nationwide, insurers that have business exposures in New York and Washington, D.C.—the two most affected locations—are arguably on the highest alert. Therefore, these insurers' adjustment of risk management around 9/11 should be most likely driven by this unexpected event. Based on

this intuition, we restrict our treatment group to only insurers that have business exposures in New York and D.C.²⁹

Panel B of Table 8 reports the results. We determine an insurer's business exposures in New York and D.C. based on whether it collects underwriting premiums from the two states during 2000—one year before the 9/11 attacks. The results confirm the previous DID findings: the increase in *RMI* for P&C insurers significantly constrains their exposure to ABS investment (as indicated by the negative coefficients of the DID estimators in Columns (1) to (3)). This effect, however, is not observed for life insurers (as shown in Columns (4) to (6)).³⁰

4.2.3 Discussions of the difference-in-differences analyses. Before we conclude this section, we would like to note that the purpose of the DID analyses is not to exclude all factors that might be correlated with insurers' ABS investment decisions. In fact, certain factors, such as insurers' risk attitude, are likely an inherent part of risk management—that is, changes in insurers' risk management are arguably elicited by changes in insurers' risk attitude. In this case, we are unable to completely tease out risk attitude from risk management. However, given that our propensity score matching (Section 3.2) and within-life-insurers analyses (Section 3.3) help mute the differences between P&C and life insurers, and that our main findings remain in a DID setting that is more resilient to unobservable confounding factors, we conclude that the differential effects of risk management on ABS investment are most likely attributable to different financial regulations insurers face rather than to confounding factors across different insurers.

4.3 Additional robustness tests

In Tables IA5 and IA6 of the Online Appendix, we present two additional robustness tests. First, we consider that insurers may have different preferences for securities with low liquidity. If ABS investments, particularly those with high underlying risk, are less liquid than other assets (e.g., Gorton 2009), and if the *RMI* measure happens to correlate with insurers' liquidity preferences, then our findings may simply suggest that financial regulations shape insurers' liquidity preferences instead of different risk management efficacy. To address this concern, we calculate the par-value weighted average of durations of all ABS held by an insurer, and repeat our analyses including duration as an additional control. Second, we use two additional measures to capture ABS

²⁹ We thank an anonymous reviewer for suggesting this test.

³⁰ It is worth noting that as two leading financial centers, New York and D.C. have fostered the fast expansion of the ABS market before the crisis. Naturally, firms with business in these two areas will be exposed to abundant ABS investment opportunities. In our context, such an exposure may partially counter *RMI*'s role in constraining ABS investment; therefore, when we restrict the treatment group to insurers with business in these two areas, we see a weaker treatment effect as compared with that in panel A of Table 8.

lurking risk (the reputation of ABS underwriters and default status of ABS). We find overall consistent results.

5. Conclusion

Following the 2008 financial crisis, regulatory proposals advocate the installation of stronger risk management to help financial institutions limit investment with substantial risk, such as ABS. In this paper, we examine this view by investigating whether financial regulations, in particular the combination of capital requirements and regulatory reporting rules, affect the efficacy of risk management in ABS investment.

Using U.S. insurance companies as a laboratory, we show that strong risk management not only lowers insurers' ABS exposure during the precrisis period but also leads them to hold ABS with higher quality. Therefore, strong risk management helps insurers identify the ABS that are masked as "safe" but subsequently suffer significant losses. However, this function of risk management is observed only among insurers facing the mark-to-market rule in interaction with capital requirements. It is considerably weaker for insurers under the historical-cost-accounting rule and capital requirements. There is suggestive evidence that the combination of mark-to-market rule and capital requirements engenders more effective risk management through promoting greater due diligence from risk managers in risk discovery and curbing executives' risk-taking incentives.

Our findings highlight that financial regulations play an important role in affecting the efficacy of risk management. To the best of our knowledge, this observation has not been previously documented in the literature. Our findings suggest that merely installing risk management in financial institutions may not be sufficient to constrain risk taking. It is crucial for regulators to consider the effects of external financial regulations and bring into play the benefits of mark-to-market accounting and capital requirements to ultimately strengthen the functioning of financial institutions' internal risk management.

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